GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Answer Key

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

96 question sets • 2915 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**HSG-CO.A.1**	Precise definitions of angle, circle, and line segment	8
Set 2	**HSG-CO.A.1**	Precise definitions of perpendicular lines and parallel lines	45
Set 3	**HSG-CO.A.2**	Transformations as functions: inputs, outputs, and preimages	76
Set 4	**HSG-CO.A.2**	Compare transformations that preserve distance and angle with those that do not	113
Set 5	**HSG-CO.A.3**	Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto it	144
Set 6	**HSG-CO.A.3**	Rotations and reflections that carry a regular polygon onto itself	181
Set 7	**HSG-CO.A.4**	Definitions of reflections and translations	218
Set 8	**HSG-CO.A.4**	Definition of a rotation	255
Set 9	**HSG-CO.A.5**	Draw the image of a figure under a rotation, reflection, or translation	292
Set 10	**HSG-CO.A.5**	Specify a sequence of transformations that carries one figure onto another	329
Set 11	**HSG-CO.B.6**	Predict the effect of a rigid motion on a figure	366
Set 12	**HSG-CO.B.6**	Decide whether two figures are congruent using rigid motions	397
Set 13	**HSG-CO.B.7**	Congruent triangles: a rigid motion exactly when all corresponding parts are congr	428
Set 14	**HSG-CO.B.8**	Explain SAS congruence with rigid motions	465
Set 15	**HSG-CO.B.8**	Explain ASA congruence with rigid motions	502
Set 16	**HSG-CO.B.8**	Explain SSS congruence with rigid motions	539
Set 17	**HSG-CO.C.9**	Prove that vertical angles are congruent	576

CONTENTS (continued)

Set 18	**HSG-CO.C.9**	Prove the angle theorems for parallel lines cut by a transversal	613
Set 19	**HSG-CO.C.9**	Prove the perpendicular bisector theorem and its converse	650
Set 20	**HSG-CO.C.10**	Prove that the angles of a triangle add to 180°	687
Set 21	**HSG-CO.C.10**	Prove that the base angles of an isosceles triangle are congruent	724
Set 22	**HSG-CO.C.10**	Prove the midsegment theorem	761
Set 23	**HSG-CO.C.10**	Prove that the medians of a triangle meet at one point	804
Set 24	**HSG-CO.C.11**	Prove that opposite sides and opposite angles of a parallelogram are congruent	847
Set 25	**HSG-CO.C.11**	Prove that the diagonals of a parallelogram bisect each other	884
Set 26	**HSG-CO.C.11**	Prove the converses: conditions that make a quadrilateral a parallelogram	921
Set 27	**HSG-CO.C.11**	Prove that rectangles are parallelograms with congruent diagonals	958
Set 28	**HSG-CO.D.12**	Construct a copy of a segment	995
Set 29	**HSG-CO.D.12**	Construct a copy of an angle	1032
Set 30	**HSG-CO.D.12**	Construct the perpendicular bisector of a segment (bisect a segment)	1069
Set 31	**HSG-CO.D.12**	Construct the bisector of an angle	1106
Set 32	**HSG-CO.D.12**	Construct a perpendicular to a line through a point on or off the line	1143
Set 33	**HSG-CO.D.12**	Construct a line parallel to a given line through a point not on it	1180
Set 34	**HSG-CO.D.13**	Construct an equilateral triangle inscribed in a circle	1217

CONTENTS (continued)

Set 35	**HSG-CO.D.13**	Construct a square inscribed in a circle	1254
Set 36	**HSG-CO.D.13**	Construct a regular hexagon inscribed in a circle	1291
Set 37	**HSG-SRT.A.1.a**	Verify that a dilation maps a line to a parallel line (or to itself through the center)	1328
Set 38	**HSG-SRT.A.1.b**	Verify that a dilation multiplies lengths by the scale factor	1365
Set 39	**HSG-SRT.A.2**	Decide whether two figures are similar using similarity transformations	1402
Set 40	**HSG-SRT.A.2**	Explain similarity of triangles as equal angles and proportional sides	1439
Set 41	**HSG-SRT.A.3**	Establish the AA criterion with similarity transformations	1476
Set 42	**HSG-SRT.B.4**	Prove the triangle proportionality theorem	1513
Set 43	**HSG-SRT.B.4**	Prove the converse of the triangle proportionality theorem	1550
Set 44	**HSG-SRT.B.4**	Prove the Pythagorean Theorem using similar triangles	1587
Set 45	**HSG-SRT.B.5**	Use similarity criteria to solve problems in geometric figures	1624
Set 46	**HSG-SRT.B.5**	Use congruence criteria to prove relationships in figures	1661
Set 47	**HSG-SRT.B.5**	Use similarity criteria to prove relationships in figures	1698
Set 48	**HSG-SRT.C.6**	Understand trigonometric ratios as properties of an angle, by similarity	1735
Set 49	**HSG-SRT.C.7**	Explain and use the relationship between the sine and cosine of complementary	1772
Set 50	**HSG-SRT.C.8**	Find unknown sides of right triangles with trigonometric ratios in applied problems	1809
Set 51	**HSG-SRT.C.8**	Find unknown angles with inverse trigonometric ratios in applied problems	1846

CONTENTS (continued)

Set 52	**HSG-SRT.C.8**	Solve right triangles in applied problems with trigonometry and the Pythagorean T	1883
Set 53	**HSG-C.A.1**	Prove that all circles are similar	1920
Set 54	**HSG-C.A.2**	Relate inscribed angles and central angles	1957
Set 55	**HSG-C.A.2**	Inscribed angles on a diameter are right angles	1994
Set 56	**HSG-C.A.2**	A radius is perpendicular to the tangent at the point of tangency	2031
Set 57	**HSG-C.A.2**	Relate circumscribed angles and central angles	2068
Set 58	**HSG-C.A.2**	Relationships among radii and chords	2105
Set 59	**HSG-C.A.3**	Construct the inscribed circle of a triangle	2142
Set 60	**HSG-C.A.3**	Construct the circumscribed circle of a triangle	2179
Set 61	**HSG-C.A.3**	Prove that opposite angles of an inscribed quadrilateral are supplementary	2216
Set 62	**HSG-C.B.5**	Derive arc length from similarity and define radian measure	2253
Set 63	**HSG-C.B.5**	Derive and use the formula for the area of a sector	2290
Set 64	**HSG-GPE.A.1**	Derive the equation of a circle	2327
Set 65	**HSG-GPE.A.1**	Complete the square to find the center and radius of a circle	2364
Set 66	**HSG-GPE.A.2**	Derive the equation of a parabola from a focus and directrix	2401
Set 67	**HSG-GPE.B.4**	Prove or disprove the type of a quadrilateral with coordinates	2444
Set 68	**HSG-GPE.B.4**	Prove or disprove that a point lies on a circle, and other coordinate proofs	2481

CONTENTS (continued)

Set 69	**HSG-GPE.B.5**	Prove the slope criteria for parallel and perpendicular lines	2518
Set 70	**HSG-GPE.B.5**	Write equations of parallel and perpendicular lines	2555
Set 71	**HSG-GPE.B.5**	Solve geometric problems with the slope criteria	2598
Set 72	**HSG-GPE.B.6**	Partition a directed segment in a given ratio	2641
Set 73	**HSG-GPE.B.7**	Compute perimeters of polygons with coordinates	2678
Set 74	**HSG-GPE.B.7**	Compute areas of triangles and rectangles with coordinates	2715
Set 75	**HSG-GMD.A.1**	Informal arguments for the circumference and area of a circle	2752
Set 76	**HSG-GMD.A.1**	Informal argument for the volume of a cylinder	2789
Set 77	**HSG-GMD.A.1**	Informal arguments for the volumes of a pyramid and a cone	2826
Set 78	**HSG-GMD.A.3**	Use volume formulas for cylinders, pyramids, and cones	2863
Set 79	**HSG-GMD.A.3**	Use the volume formula for spheres, including composite solids	2906
Set 80	**HSG-GMD.A.3**	Solve for an unknown dimension from a volume	2949
Set 81	**HSG-GMD.B.4**	Identify cross-sections of three-dimensional objects	2986
Set 82	**HSG-GMD.B.4**	Identify three-dimensional objects generated by rotations of two-dimensional obj	3023
Set 83	**HSG-MG.A.1**	Model objects with geometric shapes	3060
Set 84	**HSG-MG.A.2**	Apply density based on area	3097
Set 85	**HSG-MG.A.2**	Apply density based on volume	3134

CONTENTS (continued)

Set 86	**HSG-MG.A.3**	Design to satisfy physical constraints	3171
Set 87	**HSG-MG.A.3**	Design to minimize cost and work with typographic grids	3214
Set 88	**HSG-SRT.D.9**	Derive and use the formula A = ½ab sin C	3251
Set 89	**HSG-SRT.D.10**	Prove the Law of Sines	3288
Set 90	**HSG-SRT.D.10**	Prove the Law of Cosines	3325
Set 91	**HSG-SRT.D.11**	Apply the Law of Sines, including the ambiguous case	3368
Set 92	**HSG-SRT.D.11**	Apply the Law of Cosines, including resultant forces	3411
Set 93	**HSG-C.A.4**	Construct a tangent line from a point outside a circle	3448
Set 94	**HSG-GPE.A.3**	Derive the equation of an ellipse from its foci	3485
Set 95	**HSG-GPE.A.3**	Derive the equation of a hyperbola from its foci	3522
Set 96	**HSG-GMD.A.2**	Use Cavalieri's principle to justify the volume of a sphere	3559

Precise definitions of angle, circle, and line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
FORWARD 1	Test proposed definitions with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of new figures from earlier definitions	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Use only the words point, plane, and distance, and words built from them.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

10

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

A circle is the set of all points in a plane that are a given distance (the radius) from a given point in that plane (the center).

GRADING NOTE Any equivalent wording. Essential: (1) ALL points, (2) in one plane, (3) at one fixed distance, (4) from one fixed point, the center. Not accepted: "a round shape" (not precise); leaving out the plane (that describes a sphere); "points within a distance" (that is a disk).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Every point of line ℓ

B. Points A and B together with every point C of line ℓ for which AC + CB = AB

C. Point A and every point of line ℓ on the same side of A as B

D. Every point of the plane that is closer to A than to B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

11

QUESTION ID

S1-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Points A and B together with every point C of line ℓ for which AC + CB = AB

GRADING NOTE A segment is its two endpoints and every point between them; "C is between A and B" means C is on the line and AC + CB = AB (distance along a line). The other choices describe line ℓ, ray AB, and a half-plane.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

12

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

An angle is the figure formed by two rays that have the same endpoint. The common endpoint is the vertex, and the two rays are the sides.

GRADING NOTE Any equivalent wording. Essential: two rays (not lines), one common endpoint, that endpoint named as the vertex. Not accepted: "the space between two lines", "a turn", or "two lines that cross" (two crossing lines form four angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

13

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

She must add "in a plane" (the plane that contains O). Without it, all points 5 cm from O form a sphere, not a circle.

GRADING NOTE Any equivalent wording that adds the plane condition and explains that the points in space would form a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

True or false?

An angle with its vertex at the center of a circle that cuts off $\frac{1}{360}$ of the circle measures 1°, whatever the radius of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

14

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Degree measure is defined from distance around a circular arc: a 1° angle cuts off 1/360 of any circle centered at its vertex. The fraction of the circle, not the radius, sets the measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

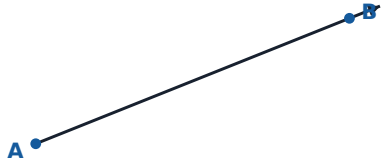
What is shown in the figure?

A. Ray AB

B. Ray BA

C. Line AB

D. Segment AB



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

16

QUESTION ID

S1-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Ray AB

GRADING NOTE It starts at A and goes through B without end, so it is ray AB (endpoint named first).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

17

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A segment has 2 endpoints; a ray has 1.

GRADING NOTE Any equivalent form. Both numbers are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

18

QUESTION ID

S1-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A segment is two endpoints and the points between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

Which figure goes on without end in exactly one direction?

A. A line

B. A point

C. A line segment

D. A ray

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

19

QUESTION ID

S1-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A ray

GRADING NOTE A ray has one endpoint and continues in one direction; a line continues in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5

20

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Ray AB starts at A and goes through B; ray BA starts at B and goes through A, in the opposite direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5

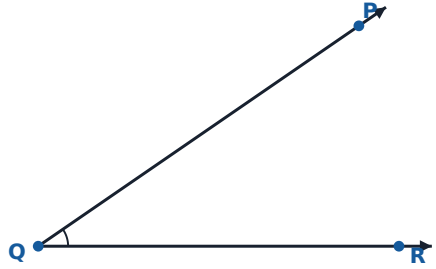
SET 1 • HSG-CO.A.1

BACKWARD 2

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

22

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Vertex Q; sides ray QP and ray QR.

GRADING NOTE Any equivalent form. The middle letter names the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

23

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. 1/4 of 360° is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

24

QUESTION ID

S1-B2-Q3

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{6}$

GRADING NOTE Any equivalent form. 60/360 = 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

25

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sides are rays; drawing them longer does not change the opening between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

26

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the meaning of one degree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

28

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Every point of the plane that is 7 cm from O is on the circle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

29

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9 in.

GRADING NOTE Any equivalent form. The radius is half the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

30

QUESTION ID

S1-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Inside the circle

GRADING NOTE D is closer to the center than the radius, so it is inside the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

31

QUESTION ID

S1-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All points of a circle are the same distance from its center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

32

QUESTION ID

S1-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the circle are exactly 5 units from the center; this point is outside.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test proposed definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

34

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

For example, two points that are each 3 cm from O (or an arc of a circle, or a sphere). The definition must say ALL points at that distance, in one plane.

GRADING NOTE Any set that fits the wording but is not a whole circle: finitely many points, an arc, or a sphere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

35

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The top vertex of an isosceles triangle with base AB has AM = MB but is not on AB. Repair: M is the point ON segment AB with AM = MB.

GRADING NOTE Any counterexample with M off segment AB and AM = MB, plus a repair that puts M on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A circle of radius 2 centered at O in a plane

B. A sphere of radius 2 centered at O

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

36

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sphere of radius 2 centered at O

GRADING NOTE In space, the points 2 units from O form a sphere. A disk also contains points closer than 2 units, so it does not fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

True or false?

"Two lines are parallel if they never meet" is a complete definition, even in three-dimensional space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

37

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Skew lines in space never meet but are not parallel. The definition needs "in the same plane".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

38

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A short segment through the center whose endpoints are not on the circle. Repair: a diameter is a segment through the center whose endpoints are both on the circle.

GRADING NOTE Any counterexample plus a repair that requires both endpoints on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of new figures from earlier definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

40

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Ray AB is point A, point B, every point between A and B, and every point P of line AB such that B is between A and P.

GRADING NOTE Any equivalent definition built from A, B, line AB, and betweenness.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

41

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The midpoint of segment AB is the point M on segment AB such that AM = MB.

GRADING NOTE Any equivalent wording. M must be on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

42

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

A semicircle is P, Q, and all points of the circle that lie on one side of line PQ.

GRADING NOTE Any equivalent wording. Essential: the endpoints of a diameter and the points of the circle on one side of the diameter line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the bisector of ∠ABC?

A. A ray from B that lies outside ∠ABC

B. The segment that joins A and C

C. Any line through B

D. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

43

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

GRADING NOTE The bisector is the ray from the vertex that splits the angle into two angles of equal measure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

The definition "the midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

44

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point on the perpendicular bisector of AB satisfies AM = MB; only one of them is on the segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definitions of perpendicular lines and parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular and parallel lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about three lines in a plane and in space	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

47

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

Two lines are parallel if they lie in the same plane and do not intersect.

GRADING NOTE Any equivalent wording. Essential: in the same plane, and no point in common. Not accepted: "lines that never meet" without the plane (skew lines in space never meet), or "lines that point the same way" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

48

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

Two lines are perpendicular if they intersect and form a right angle.

GRADING NOTE Any equivalent wording, such as "intersect to form four right angles" or "intersect to form congruent adjacent angles". Not accepted: "lines that cross" or "lines that make a T" (not precise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

49

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They lie in the same plane

GRADING NOTE Parallel lines are coplanar lines that do not meet. Lines in space can miss each other without being parallel (skew lines), even if both are perpendicular to one segment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

True or false?

If two lines meet and form one right angle, they form four right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

50

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The angle next to a right angle is its supplement, 90°, and vertical angles are equal, so all four angles are 90°.

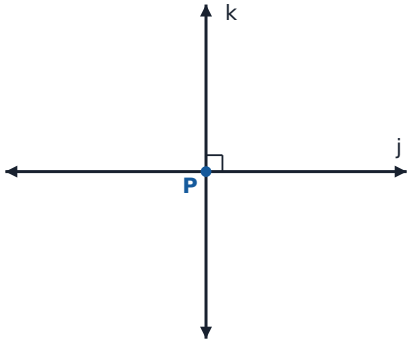
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

51

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. The lines intersect and form a right angle, which is the definition of perpendicular lines.

GRADING NOTE Any equivalent explanation that names both parts of the definition: they intersect, and an angle is a right angle (given by the mark).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular and parallel lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

53

QUESTION ID

S2-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of perpendicular lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

54

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Lines can cross at angles other than 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

55

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

56

QUESTION ID

S2-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. They are parallel

GRADING NOTE Opposite sides of a rectangle never meet and lie in one plane, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two straight streets that never meet

B. Two streets that curve around a park

C. Two streets that meet at a sharp point

D. Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

57

QUESTION ID

S2-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Two streets that cross at a square corner

GRADING NOTE Perpendicular lines meet at a right (square) angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

59

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

90°, 90°, 90°

GRADING NOTE Any order. The adjacent angles are 180° − 90°; the vertical angle is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

60

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

110°, 70°, 110°

GRADING NOTE Any order. Adjacent angles: 180° − 70° = 110°; the vertical angle is 70°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

61

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical angles have equal measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 180°

B. 45°

C. 90°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

62

QUESTION ID

S2-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 90°

GRADING NOTE A linear pair adds to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°.
The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

63

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The angle next to it forms a linear pair with it, so it measures 80°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P. Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

65

QUESTION ID

S2-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four angles measure 90°.

GRADING NOTE REQUIRED WORK: Start from the definition (one angle at P is a right angle). Use the linear pair (supplementary) to get the adjacent angles and vertical angles (or another linear pair) to get the fourth, with a reason for each step.

Any valid proof. Essential: (1) definition gives one 90° angle; (2) each adjacent angle forms a linear pair with it, so it is 180° − 90° = 90°; (3) the fourth angle is vertical to the first (or forms a linear pair with a 90° angle). Invalid: "they look like right angles".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

66

QUESTION ID

S2-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Each angle of the pair measures 90°, so the lines are perpendicular.

GRADING NOTE REQUIRED WORK: Linear pair angles add to 180°; congruent means equal measures, so each is 90°; then apply the definition.

Any valid proof. Essential: x + x = 180° from the linear pair, so x = 90°; the lines meet and form a right angle, which is the definition of perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

67

QUESTION ID

S2-F1-Q3

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form. Perpendicular means the angle is 90°: 3x + 15 = 90, so x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

True or false?

If two intersecting lines form a pair of vertical angles that are supplementary, the lines are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

68

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical angles are equal, so each is 180° ÷ 2 = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at point O. Which statement is enough to prove that p ⊥ q?

A. Two of the angles at O are vertical angles

B. Two adjacent angles at O add to 180°

C. Two of the angles at O are acute

D. One of the four angles at O measures 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

69

QUESTION ID

S2-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. One of the four angles at O measures 90°

GRADING NOTE The definition needs a right angle. Every pair of crossing lines has vertical angles and linear pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about three lines in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

71

QUESTION ID

S2-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

a ∥ b

GRADING NOTE REQUIRED WORK: Use the corresponding angles that a and b make with t (both right angles, so congruent) and the converse of the corresponding angles theorem; or argue by contradiction with the triangle angle sum.

Any valid proof. Essential: both corresponding angles at t are 90°, so they are congruent; congruent corresponding angles make the lines parallel. Alternative: if a and b met, they would form a triangle with t having two 90° angles, so its angle sum would exceed 180°. Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

72

QUESTION ID

S2-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Counterexample: in a corner of a room, the two floor edges are both perpendicular to the vertical edge, but they meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is perpendicular to b

C. t is parallel to b

D. The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

73

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. t is perpendicular to b

GRADING NOTE t crosses a, so it crosses b. The corresponding angle at b equals the right angle at a, so t ⊥ b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

74

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

x = 5

GRADING NOTE Form required: an equation of the line. A line perpendicular to the x-axis is vertical, so every point has x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

75

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the parallel postulate of Euclidean geometry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

78

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 2)

GRADING NOTE Form required: an ordered pair. Input (2, 5): x + 4 = 6 and y − 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by R(x, y) = (−y, x).
Find R(3, −1).

R(3, −1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

79

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 3)

GRADING NOTE Form required: an ordered pair. −y = −(−1) = 1 and x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

80

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

(4, −7)

GRADING NOTE Form required: an ordered pair. F(4, −7) = (4, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. T(x, y) = (x − 5, y − 2)

B. T(x, y) = (5x, 2y)

C. T(x, y) = (x + 2, y + 5)

D. T(x, y) = (x + 5, y + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

81

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. T(x, y) = (x + 5, y + 2)

GRADING NOTE Right adds to x and up adds to y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

True or false?

Under the transformation D(x, y) = (2x, 2y), the point (0, 0) is its own output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

82

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE D(0, 0) = (0, 0): the origin is a fixed point of D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 3x − 2. Find f(4).

f(4) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

84

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. 3(4) − 2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

85

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

g(x) = x + 5. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

86

QUESTION ID

S3-B1-Q3

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. 7 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

87

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

88

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A function gives exactly one output for each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

90

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

91

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (4, −1)

D. (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

92

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−4, −1)

GRADING NOTE A 180° rotation about the origin sends (x, y) to (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

93

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

94

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (x, y) → (−y, x) gives (−5, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

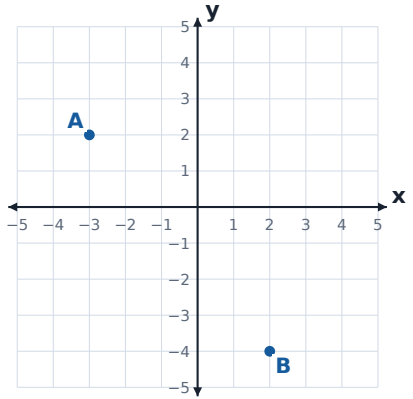
SET 3 • HSG-CO.A.2

BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

96

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

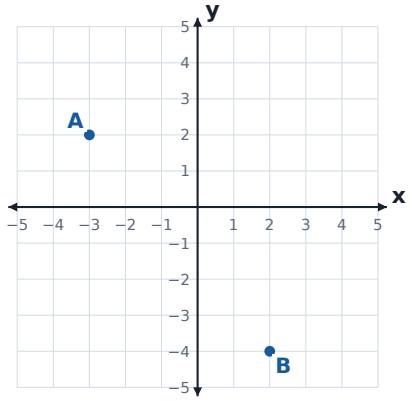
GRADING NOTE Form required: an ordered pair (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

97

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −4)

GRADING NOTE Form required: an ordered pair (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. (−2, −5)

B. (−2, 5)

C. (2, 5)

D. (2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

98

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (−2, −5)

GRADING NOTE Quadrant III has $x < 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

99

QUESTION ID

S3-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its x-coordinate is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point (−2, 5) is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

100

QUESTION ID

S3-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = −2 is 2 units to the LEFT of the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

T(x, y) = (x + 2, y) and R(x, y) = (−x, y). Find R(T(1, 4)).

R(T(1, 4)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

102

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, 4)

GRADING NOTE Form required: an ordered pair. T(1, 4) = (3, 4); R(3, 4) = (−3, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

103

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−x − 2, y)

GRADING NOTE Any equivalent form, such as (−(x + 2), y). Apply T first, then R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $T(R(1, 4))$.

$T(R(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

104

QUESTION ID

S3-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 4)

GRADING NOTE Form required: an ordered pair. $R(1, 4) = (-1, 4)$; $T(-1, 4) = (1, 4)$. The order changes the answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

105

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation and a reflection give different results in different orders (for example, R(T(1, 4)) ≠ T(R(1, 4)) above).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$. Which rule is $S(T(x, y))$?

A. $(x + 2, y + 4)$

B. $(x - 3, y - 5)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

106

QUESTION ID

S3-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. (x + 2, y + 4)

GRADING NOTE $S(x + 3, y - 1) = (x + 3 - 1, y - 1 + 5) = (x + 2, y + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of T(x, y) = (x − 6, y + 2).

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

108

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6, y − 2)

GRADING NOTE Any equivalent form. The inverse undoes T: T(x + 6, y − 2) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

R(x, y) = (−y, x) is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

109

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(y, −x)

GRADING NOTE Any equivalent form: a 90° clockwise rotation. Check: R(y, −x) = (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every rotation is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

110

QUESTION ID

S3-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 90° rotation is undone by a 270° (or −90°) rotation; only a 180° rotation undoes itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. (x − 3, y − 3)

B. (3y, 3x)

C. (−3x, −3y)

D. $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

111

QUESTION ID

S3-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (x/3, y/3)

GRADING NOTE Dividing each coordinate by 3 undoes multiplying by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

112

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P is not one-to-one: (2, 1) and (2, 5) have the same output (2, 0), so no rule can undo it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio...	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev...	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. R(x, y) = (−x, y)

B. T(x, y) = (x + 4, y − 1)

C. H(x, y) = (2x, y)

D. Q(x, y) = (−y, x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

115

QUESTION ID

S4-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. H(x, y) = (2x, y)

GRADING NOTE H stretches horizontally: (0, 0) and (1, 0) are 1 unit apart, but their images (0, 0) and (2, 0) are 2 units apart. The others are a translation, a reflection, and a rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

116

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

H(A) = (0, 0), H(B) = (6, 4); the distance is √52 = 2√13 ≈ 7.21; no.

GRADING NOTE Form required: both images, the distance (exact or to 0.01), and the conclusion. $\sqrt{(6^2 + 4^2)} = \sqrt{52} \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation T(x, y) = (x − 2, y + 5) maps A(1, 1) and B(4, 5) to A′ and B′. Show that A′B′ = AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

117

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

A′(−1, 6), B′(2, 10).
AB = √(3² + 4²) = 5
and
A′B′ = √(3² + 4²) = 5,
so A′B′ = AB.

GRADING NOTE Form required: both images and both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

True or false?

The transformation V(x, y) = (x, 3y) preserves the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

118

QUESTION ID

S4-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The right angle at (0, 0) between the rays through (1, 1) and (1, −1) maps to the angle between the rays through (1, 3) and (1, −3), which is about 143°, not 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

119

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

Distance: no; every distance is multiplied by 3. Angle measure: yes; S is a dilation, which keeps every angle measure.

GRADING NOTE Form required: both answers with reasons. Example: (0, 0) and (1, 0) are 1 apart; their images are 3 apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

121

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between (−2, 3) and (4, 3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

122

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. Same y, so |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

123

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. √(25 + 144) = 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

124

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(1^2 + 1^2)} = \sqrt{2} \approx 1.41$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between (2, −1) and (5, 3).

A. 25

B. √7

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

125

QUESTION ID

S4-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 5

GRADING NOTE $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

127

QUESTION ID

S4-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

128

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. Its side lengths

C. The positions of its vertices

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

129

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The positions of its vertices

GRADING NOTE A translation moves the figure but keeps its lengths, angles, and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

130

QUESTION ID

S4-B2-Q4

TYPE

Short answer

CORRECT ANSWER

7 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

131

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

35°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. (x, y) → (4x, y)

B. (x, y) → (x + 4, y)

C. (x, y) → (x, y + 4)

D. (x, y) → (4x, 4y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

133

QUESTION ID

S4-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (4x, 4y)

GRADING NOTE Multiplying both coordinates by 4 is a dilation: angles stay, distances are multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify (x, y) → (−2x, −2y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

134

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A similarity transformation that is not rigid: every distance is doubled and angles are kept (a dilation by 2 followed by a 180° rotation).

GRADING NOTE Any equivalent explanation that shows the factor 2 for distances and that angles are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify (x, y) → (x + y, y) as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

135

QUESTION ID

S4-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Neither. (0, 0) and (0, 1) are 1 apart but their images (0, 0) and (1, 1) are √2 apart, while (0, 0) and (1, 0) stay 1 apart: distances are not all multiplied by one number, and the right angle at the origin becomes 45°.

GRADING NOTE Any correct counterexample showing that distances are not scaled by a single factor (or an angle changes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every similarity transformation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

136

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only similarity transformations with scale factor 1 are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

137

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiplying all distances by k keeps triangles similar (SSS), so angle measures stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that T(x, y) = (x + a, y + b) preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

139

QUESTION ID

S4-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The image distance equals $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$, the original distance.

GRADING NOTE

REQUIRED WORK: Write both images, apply the distance formula to the images, and show that a and b cancel.

Any valid algebraic proof. Essential: images $(x_1 + a, y_1 + b)$ and $(x_2 + a, y_2 + b)$; the differences are $x_2 - x_1$ and $y_2 - y_1$, so the distance formula gives the same value. Invalid: checking one or two numerical examples only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

140

QUESTION ID

S4-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

The image distance is $3\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

GRADING NOTE REQUIRED WORK: Write both images, apply the distance formula, and factor out 9 under the square root.

Any valid proof. Essential: $\sqrt{((3x_2 - 3x_1)^2 + (3y_2 - 3y_1)^2)} = \sqrt{(9[(x_2 - x_1)^2 + (y_2 - y_1)^2])} = 3\cdot$(original distance).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

141

QUESTION ID

S4-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Exactly the pairs with $x_1 = x_2$ (points on the same vertical line).

GRADING NOTE Any equivalent condition. The distance is unchanged only if $4(x_2 - x_1)^2 = (x_2 - x_1)^2$, that is, $x_2 = x_1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

142

QUESTION ID
S4-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE H(x, y) = (2x, y) keeps vertical distances but changes others. A general claim needs a proof for all points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

One example in which a distance is preserved is enough to prove that a transformation preserves distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

143

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every pair of points must keep its distance; one example proves nothing about all pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

146

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rotations: 180° and 360° about the center O. Reflections: across the line through the midpoints of AB and CD, and across the line through the midpoints of BC and DA.

GRADING NOTE Form required: both rotations and both reflections, with the center and the lines named. The diagonals are NOT lines of symmetry of a non-square rectangle, and a 90° turn does not work because the sides have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Which transformation carries it onto itself?

A. A 90° rotation about the point where its diagonals meet

B. A reflection across diagonal PR

C. A reflection across the line through the midpoints of PQ and RS

D. A 180° rotation about the point where its diagonals meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

147

QUESTION ID

S5-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

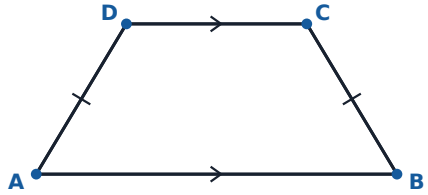
D. A 180° rotation about the point where its diagonals meet

GRADING NOTE The diagonals of a parallelogram bisect each other, so a 180° turn about their intersection swaps P with R and Q with S. A general parallelogram has no line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD is isosceles, with bases AB and DC and legs AD = BC. Describe every rotation and reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

148

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

Only the reflection across the line through the midpoints of the two bases, and the 360° rotation.

GRADING NOTE Form required: the one reflection (named by its line) and the 360° rotation. No rotation of less than 360° works because the bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

True or false?

A 90° rotation about its center carries a rectangle that is not a square onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

149

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 90° turn would send a long side onto a short side, so the image does not match the rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has EF ∥ HG, EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations and reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

150

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

Only the 360° rotation (the identity). It has no line of symmetry and no other rotation symmetry.

GRADING NOTE Form required: the conclusion with a reason. A symmetry would have to send the 3-unit leg to a congruent leg, but the only other leg is 5 units long, and the two bases have different lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

152

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. Two through opposite vertices and two through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

153

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Only the lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

154

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Folding along a diagonal does not match the sides of a non-square rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. An isosceles triangle that is not equilateral

B. A square

C. A circle

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

155

QUESTION ID

S5-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. An isosceles triangle that is not equilateral

GRADING NOTE The line through the vertex angle and the midpoint of the base is its only line of symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

156

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The line through the midpoints of its two bases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

158

QUESTION ID

S5-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only rectangles (including squares) have equal diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

159

QUESTION ID

S5-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both pairs of opposite sides of a rectangle are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

160

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Four right angles

GRADING NOTE A rectangle has four right angles; only a square has all sides equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

161

QUESTION ID

S5-B2-Q4

TYPE

Short answer

CORRECT ANSWER

110°, 70°, 110°

GRADING NOTE Any order. Consecutive angles are supplementary; opposite angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

162

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form. Base angles of an isosceles trapezoid are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

164

QUESTION ID

S5-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −1)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

165

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 2)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

166

QUESTION ID

S5-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the 180° rotation rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point (−4, 2) by 90° counterclockwise about the origin.

A. (4, −2)

B. (−4, −2)

C. (2, 4)

D. (−2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

167

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, −4)

GRADING NOTE (x, y) → (−y, x) gives (−2, −4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

168

QUESTION ID

S5-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 360° rotation returns every point to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

170

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A goes to B (and D goes to C), so ∠DAB ≅ ∠CBA.

GRADING NOTE Any equivalent answer naming A → B and a pair of base angles (∠D ≅ ∠C also accepted). A reflection keeps angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

171

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

PQ ≅ RS and QR ≅ SP.

GRADING NOTE Any equivalent form. The rotation sends segment PQ onto segment RS and QR onto SP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

172

QUESTION ID

S5-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The reflection that swaps A with B and D with C sends diagonal AC onto diagonal BD, and reflections keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AB = BC

B. AC = AB

C. AD = AB

D. AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

173

QUESTION ID

S5-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. AD = BC

GRADING NOTE The reflection sends side AD onto side BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

174

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every parallelogram has 180° rotation symmetry; most are not rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

176

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about O.

GRADING NOTE Any equivalent form. Reflections across perpendicular lines through O combine to a 180° rotation about O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. A rectangle that is not a square

D. A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

177

QUESTION ID

S5-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. A kite that is not a rhombus

GRADING NOTE A kite has one line of symmetry, through two opposite vertices. A rectangle has two lines and a 180° turn; a general parallelogram has a 180° turn and no line; a square has eight symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

178

QUESTION ID

S5-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every parallelogram has 180° rotation symmetry about the crossing point of its diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

179

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

A parallelogram that is not a rectangle or a rhombus.

GRADING NOTE Any equivalent answer. A rectangle or a rhombus would also have reflection symmetries.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

180

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point goes to its image and back again.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

183

QUESTION ID
S6-M-Q1

TYPE
Short answer

CORRECT ANSWER
72°, 144°, 216°, 288°, and 360°

GRADING NOTE Form required: all five angles. Each is a multiple of 360° ÷ 5 = 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

184

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 reflections: 3 across lines through pairs of opposite vertices and 3 across lines through the midpoints of pairs of opposite sides.

GRADING NOTE Form required: the number and both kinds of lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Which rotation about its center carries a regular octagon onto itself?

A. 100°

B. 135°

C. 150°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

185

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 135°

GRADING NOTE The rotations are multiples of 360° ÷ 8 = 45°; 135° = 3 × 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

True or false?

Each line of symmetry of a regular nonagon (9 sides) passes through one vertex and the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

186

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A regular polygon with an odd number of sides has n lines of symmetry, each through a vertex and the midpoint of the opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

187

QUESTION ID

S6-M-Q5

TYPE

Short answer

CORRECT ANSWER

3 reflections: across each line through a vertex and the midpoint of the opposite side.

GRADING NOTE Form required: the number and the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

189

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

72°

GRADING NOTE Any equivalent form. 360 ÷ 5 = 72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

190

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form. 360 ÷ 8 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

191

QUESTION ID

S6-B1-Q3

TYPE

Short answer

CORRECT ANSWER

180°

GRADING NOTE Any equivalent form. 3/6 = 1/2 and 1/2 of 360 is 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

192

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 360 ÷ 12 = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

193

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360 ÷ 9 = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. A rhombus that is not a square

C. A rectangle that is not a square

D. An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

195

QUESTION ID

S6-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. An equilateral triangle

GRADING NOTE A regular polygon has all sides equal and all angles equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

196

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A rhombus that is not a square has equal sides but unequal angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

197

QUESTION ID

S6-B2-Q3

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

198

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

A square

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

199

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Regular means all sides and all angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

201

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

202

QUESTION ID

S6-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

203

QUESTION ID

S6-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A reflection that carries the polygon onto itself keeps its center fixed, so the center is on the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. exactly one vertex and no midpoint

C. no point of the square

D. two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

204

QUESTION ID

S6-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. two opposite vertices, or the midpoints of two opposite sides

GRADING NOTE A square has 2 diagonal lines and 2 lines through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

205

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 6: 3 through opposite vertices and 3 through midpoints of opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

207

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

208

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

18°

GRADING NOTE Any equivalent form. A regular n-gon has n lines of symmetry, so n = 20 and 360 ÷ 20 = 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

209

QUESTION ID

S6-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 360 ÷ 50 = 7.2 is not a whole number of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

210

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 40°

GRADING NOTE The angle must be 360° ÷ n for a whole number n ≥ 3; 360 ÷ 40 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

211

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. 12 rotations (30°, 60°, ..., 360°) and 12 reflections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

213

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The position of C

GRADING NOTE Any equivalent wording. Each 60° turn moves a vertex one place counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

214

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The position of E

GRADING NOTE Any equivalent wording. The rotation moves B to the position of C; the reflection across AD swaps C and E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

215

QUESTION ID

S6-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two reflections across lines through the center combine into a rotation about the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of C

B. The position of A

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The position of A

GRADING NOTE The reflection across AC sends B to the position of D; the 90° turn moves D to the position of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

217

QUESTION ID

S6-F2-Q5

TYPE

Short answer

CORRECT ANSWER

A 144° rotation about the center

GRADING NOTE Any equivalent form. Rotation angles about the same center add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definitions of reflections and translations

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular and parallel segments on a grid	**4.G.A.1**
BACKWARD 3	Reflect and translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definitions that reflections and translations keep l…	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

220

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

ℓ is the perpendicular bisector of segment PP′; itself.

GRADING NOTE Any equivalent wording. Both blanks are needed. "P′ is the same distance from ℓ" alone is not enough: it must be on the perpendicular through P, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Points A and B are different. Write a precise definition of the translation that maps A to B.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

221

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

It maps each point P to the point P′ such that PP′ has the same length as AB, is parallel to AB (or on line AB), and points in the same direction.

GRADING NOTE Any equivalent wording. Essential: same length, parallel (or collinear), same direction. Accept "ABP′P is a parallelogram" for points P not on line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

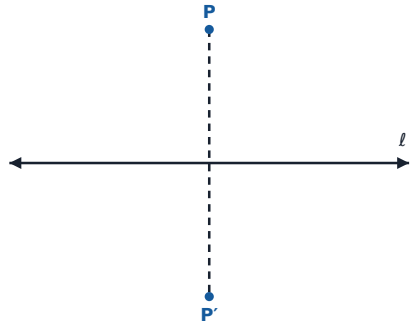
Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?

A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

222

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ℓ is perpendicular to PP′ and passes through its midpoint

GRADING NOTE By the definition of a reflection, ℓ is the perpendicular bisector of PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

223

QUESTION ID

S7-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is what the definition of a translation says about every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

224

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

x = 3

GRADING NOTE Form required: an equation of the line. The line is the perpendicular bisector of PP′: PP′ is horizontal with midpoint (3, 2), so the line is the vertical line x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

226

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. Halfway from x = 2 to x = 8 is x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

227

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

228

QUESTION ID

S7-B1-Q3

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (−3 + 7) ÷ 2 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

229

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (−4 + 6) ÷ 2 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

230

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is the same distance from both endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular and parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal segment and a vertical segment are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

232

QUESTION ID

S7-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Horizontal and vertical directions meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

233

QUESTION ID

S7-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Length does not decide direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

D. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

234

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

GRADING NOTE Both segments in the first pair go 3 right and 1 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) that is perpendicular to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

235

QUESTION ID

S7-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Any horizontal segment through (5, 3), for example from (4, 3) to (8, 3).

GRADING NOTE Any horizontal segment containing (5, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel

B. They are perpendicular to each other

C. They meet at the origin

D. They are the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

236

QUESTION ID

S7-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. They are parallel

GRADING NOTE Both are vertical lines, so they are parallel (or the same line if they have the same x-value).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect and translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

238

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Translate the point (−1, 5) 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

239

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

240

QUESTION ID

S7-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (7, 2)

GRADING NOTE Reflecting across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point (−6, 1) across the y-axis gives (6, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

241

QUESTION ID

S7-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

242

QUESTION ID

S7-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Left changes x: (1, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

244

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = -2x + 5$

GRADING NOTE Any equivalent equation. The midpoint of PP′ is (2, 1); PP′ has slope 1/2, so the perpendicular bisector has slope −2: $y - 1 = -2(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A translation maps A(2, −1) to A′(−3, 4). Where does it map B(0, 6)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

245

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 11)

GRADING NOTE Form required: an ordered pair. The translation moves every point 5 left and 5 up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

246

QUESTION ID
S7-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both segments are perpendicular to the line of reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

247

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. The midpoint (2, 2) is on y = x, and AA′ (slope −1) is perpendicular to y = x (slope 1).

GRADING NOTE Form required: the conclusion with both conditions of the definition checked (midpoint and perpendicularity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (5, 0)

C. (2, −3)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

248

QUESTION ID

S7-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. (2, −3)

GRADING NOTE The line is x = 2, the vertical perpendicular bisector of the horizontal segment PP′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definitions that reflections and translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

250

QUESTION ID

S7-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ = P′Q′

GRADING NOTE REQUIRED WORK: Use the definition (ℓ is the perpendicular bisector of PP′ and of QQ′), then two pairs of congruent triangles (SAS): PMN ≅ P′MN, then PNQ ≅ P′NQ′.

Any valid proof. Essential: PM = P′M and QN = Q′N with right angles at M and N (definition); triangles PMN and P′MN are congruent by SAS, so PN = P′N and ∠PNM ≅ ∠P′NM; subtracting from the right angles gives ∠PNQ ≅ ∠P′NQ′; then triangles PNQ and P′NQ′ are congruent by SAS, so PQ = P′Q′. Alternative: coordinates with ℓ as the x-axis. Invalid: "reflections preserve distance" (that is what is being proved).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

251

QUESTION ID

S7-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

PQ = P′Q′

GRADING NOTE REQUIRED WORK: Use the definition (PP′ and QQ′ are parallel, the same length, and the same direction) to show PP′Q′Q is a parallelogram.

Any valid proof. Essential: PP′ ∥ QQ′ and PP′ = QQ′ in the same direction, so PP′Q′Q is a parallelogram (one pair of opposite sides parallel and congruent); opposite sides of a parallelogram are congruent, so PQ = P′Q′. Invalid: assuming translations preserve length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

252

QUESTION ID

S7-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE If Q is on line PP′, all four points lie on one line, so there is no parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

253

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are on opposite sides; the line is the perpendicular bisector of the segment joining them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ is the perpendicular bisector of PP′ and of QQ′

B. ℓ bisects ∠PQP′

C. ℓ is parallel to PQ

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

254

QUESTION ID
S7-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. ℓ is the perpendicular bisector of PP′ and of QQ′

GRADING NOTE The proof starts from the definition of a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point $P \neq O$ to the point P' such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

257

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

$OP' = OP$ (P' is on the circle with center O through P); $m\angle POP' = \theta$, measured counterclockwise from ray OP; itself.

GRADING NOTE Any equivalent wording. All three blanks are needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is the midpoint of PP′

B. O is on the perpendicular bisector of PP′

C. OP′ = 2 · OP

D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

258

QUESTION ID

S8-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. O is on the perpendicular bisector of PP′

GRADING NOTE OP = OP′, so O is equidistant from P and P′ and lies on the perpendicular bisector of PP′. O is the midpoint only for a 180° rotation.

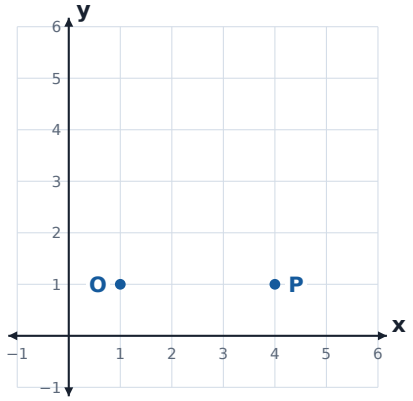
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.

P′ = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

259

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

(1, 4)

GRADING NOTE Form required: an ordered pair. P is 3 units right of O; after a 90° counterclockwise turn it is 3 units above O.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

True or false?

Under a rotation about point O, each point P moves along the circle with center O that passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

260

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The definition keeps OP′ = OP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

261

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

The center is the same distance from P and from P′, so it is on the perpendicular bisector of PP′. That bisector passes through (1, 1) with slope 1, which is the line y = x.

GRADING NOTE Any equivalent explanation that uses equal distances from the center (the definition) and the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

263

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Up

GRADING NOTE Any equivalent wording (north).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

264

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Down

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

265

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 270° clockwise and 90° counterclockwise add to a full 360° turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

266

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Right

GRADING NOTE Clockwise from up is toward the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

267

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

270°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

269

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

270

QUESTION ID

S8-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are on the circle with center O and radius OA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

271

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 4)

GRADING NOTE $\sqrt{(3^2 + 4^2)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

272

QUESTION ID

S8-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

273

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Many points lie on the circle of that radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

275

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 5)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate (−3, 4) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

276

QUESTION ID

S8-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −4)

GRADING NOTE Form required: an ordered pair. (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (1, −6)

C. (−1, −6)

D. (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

277

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, −1)

GRADING NOTE (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

278

QUESTION ID

S8-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 270° counterclockwise = 90° clockwise: (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

279

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives (0, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

281

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 0)

GRADING NOTE Form required: an ordered pair. P is (3, 1) from C; a 90° clockwise turn gives (1, −3) from C, which is (3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

282

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−3, −1)

GRADING NOTE Form required: an ordered pair. P is (2, 3) from C; the image is (−2, −3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

283

QUESTION ID

S8-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Form required: an ordered pair. P is (3, 0) from C; the image is (0, 3) from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 4)

B. (4, 1)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

284

QUESTION ID

S8-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 1)

GRADING NOTE P is (0, 3) from C; 270° counterclockwise (= 90° clockwise) gives (3, 0) from C, which is (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

285

QUESTION ID
S8-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About C the image is (2, 2); about the origin it is (0, 4). The center matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

287

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Center (0, 0); 90° counterclockwise.

GRADING NOTE Form required: the center and the angle with direction. The bisector of BB′ is y = x; the bisector of AA′ passes through (1, 2) with slope 2, which is y = 2x. They meet at (0, 0), and (1, 0) → (0, 1) is a 90° counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

288

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Center (2, 1); 90° counterclockwise.

GRADING NOTE Form required: the center and the angle. Check: P is (2, 0) from (2, 1) and P′ is (0, 2) from it; Q is (2, 2) from it and Q′ is (−2, 2) from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a nonzero rotation moves to a new position under the rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

289

QUESTION ID

S8-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The center is the one fixed point of the rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rotation about C maps P to P′. CP = 5 and PP′ = 5√2. Which could be the angle of rotation?

A. 60°

B. 90°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

290

QUESTION ID

S8-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 90°

GRADING NOTE CP = CP′ = 5 and $PP'^2 = 50 = 5^2 + 5^2$, so triangle PCP′ has a right angle at C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

291

QUESTION ID

S8-F2-Q5

TYPE

Short answer

CORRECT ANSWER

53°

GRADING NOTE Rounding to the nearest degree required. The angle at the origin has tangent 4/3, so it is about 53.13°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

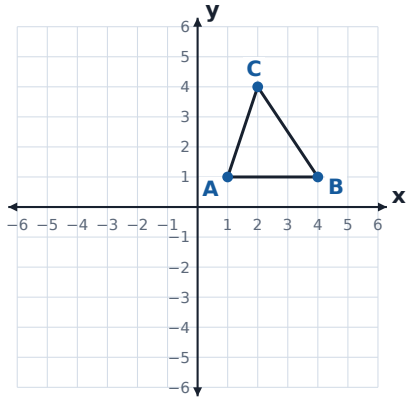
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

294

QUESTION ID

S9-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

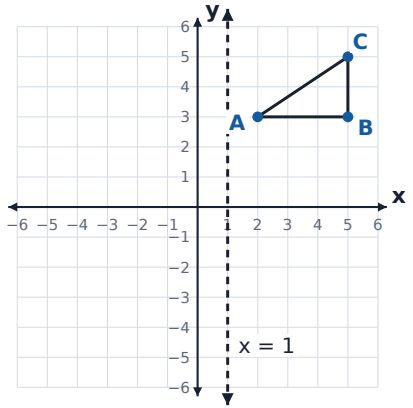
A′(−4, −1), B′(−1, −1), C′(−3, 2), joined to form triangle A′B′C′.

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (x − 5, y − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

295

QUESTION ID

S9-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

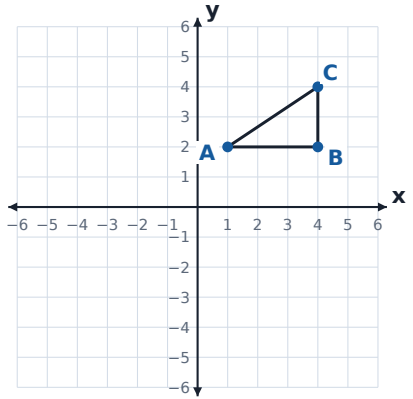
A′(0, 3), B′(−3, 3), C′(−3, 5).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is the same distance from x = 1 on the other side: x → 2 − x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

296

QUESTION ID

S9-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

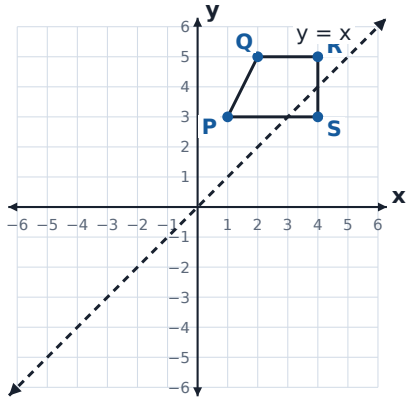
A′(−2, 1), B′(−2, 4), C′(−4, 4).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

Draw the image of quadrilateral PQRS under the reflection across the line y = x. Label the image P′Q′R′S′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

297

QUESTION ID

S9-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

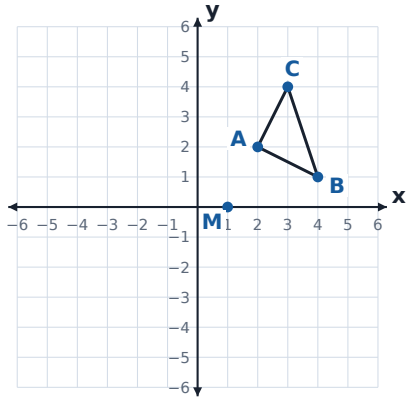
P′(3, 1), Q′(5, 2), R′(5, 4), S′(3, 4).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Reflection across y = x swaps the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

298

QUESTION ID

S9-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A′(0, −2), B′(−2, −1), C′(−1, −4).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. Each image is directly opposite its point through M: (x, y) → (2 − x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

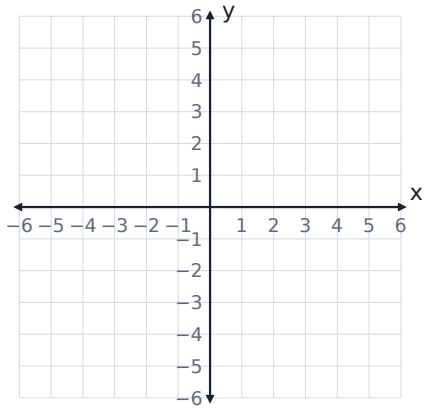
SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

ANSWER KEY Reduced copy of the question page, with its answer

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

300

QUESTION ID

S9-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A at (−3, 2), B at (0, −4), C at (5, 1).

GRADING NOTE Each point within about a quarter of a grid square of its location.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

301

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, −2)

C. (0, 2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

302

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (0, 2)

GRADING NOTE On the y-axis, x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

303

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

304

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The x-coordinate (2) is the horizontal move.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of (3, −2) under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

306

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, −1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of (−2, 5) under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

307

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (1, −3)

D. (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

308

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, −3)

GRADING NOTE (x, y) → (y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of (4, −1) under a reflection across the y-axis is (−4, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

309

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

310

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (−2, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

312

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. |5 − 1| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point (−2, −1) from the horizontal line y = 3?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

313

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. |−1 − 3| = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (−1, 4)

C. (2, 1)

D. (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

314

QUESTION ID

S9-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

GRADING NOTE 2 − 3 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

315

QUESTION ID

S9-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its distance from y = 0 is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

316

QUESTION ID

S9-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is |6 − 2| = 4 units away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B3-Q5

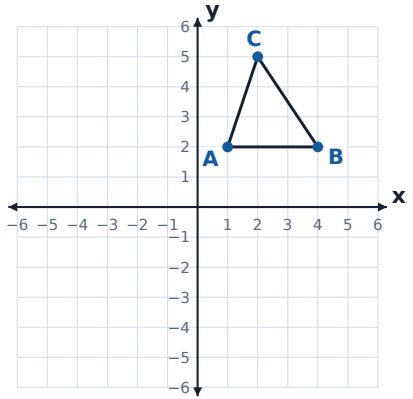
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

318

QUESTION ID

S9-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

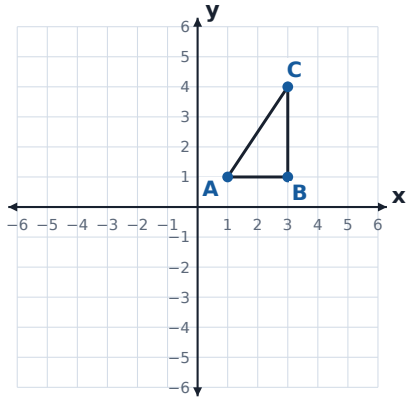
A″(−1, −1), B″(−4, −1), C″(−2, 2).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−x, y) → (−x, y − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

319

QUESTION ID

S9-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A″(−1, −1), B″(−1, −3), C″(−4, −3).

GRADING NOTE Any drawing with these vertices (within about a quarter of a grid square), joined in order and labeled. (x, y) → (−y, x) → (−y, −x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

320

QUESTION ID

S9-F1-Q3

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair. (2, 1) → (4, 1) → (−4, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

321

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two reflections give (x, y) → (−x, −y), a 180° rotation; reflecting across y = x gives (y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

322

QUESTION ID

S9-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (3, −1)

GRADING NOTE (1, 3) → (−1, 3) → (3, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

QUESTION ID
S9-F2-Q1

TYPE
Drawing / plotting

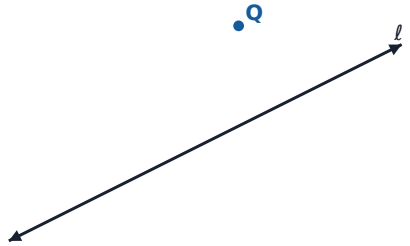
CORRECT ANSWER

P′ is on the circle with center O through P, with m∠POP′ = 60° measured counterclockwise from ray OP.

GRADING NOTE Accept within about 2° and 2 mm. Evidence: the arc (or equal compass distance) from O and the 60° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

325

QUESTION ID

S9-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

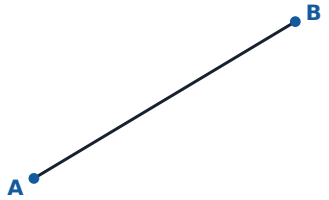
Q′ is on the perpendicular from Q to ℓ, on the other side of ℓ, the same distance from ℓ as Q (ℓ is the perpendicular bisector of QQ′).

GRADING NOTE Accept within about 2 mm. Evidence: the construction of the perpendicular from Q to ℓ and equal distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

326

QUESTION ID

S9-F2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

AB′ has the same length as AB, with m∠BAB′ = 90° measured clockwise from ray AB; A stays fixed.

GRADING NOTE Accept within about 2° and 2 mm. Evidence: equal lengths and the 90° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

True or false?

To rotate point P about point O by 45°, it is enough to draw a 45° angle at O; the distance from O does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

327

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image must also be the same distance from O as P (on the circle centered at O through P).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Describe how to reflect point P across a slanted line ℓ using only a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

328

QUESTION ID

S9-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Construct the line through P perpendicular to ℓ; mark where it meets ℓ (M); with the compass set to MP, mark the point P′ on that perpendicular, on the other side of ℓ, so that MP′ = MP.

GRADING NOTE Any equivalent construction (for example, two circles centered at two points of ℓ through P meet again at P′).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

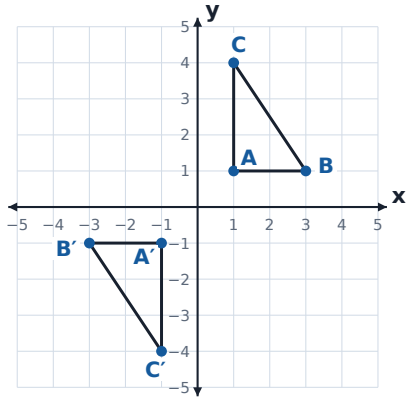
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

331

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

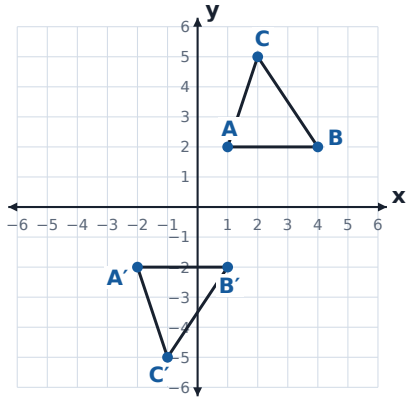
A 180° rotation about the origin.

GRADING NOTE Any valid sequence, for example a reflection across the x-axis followed by a reflection across the y-axis. Check: every vertex must land on its partner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

332

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

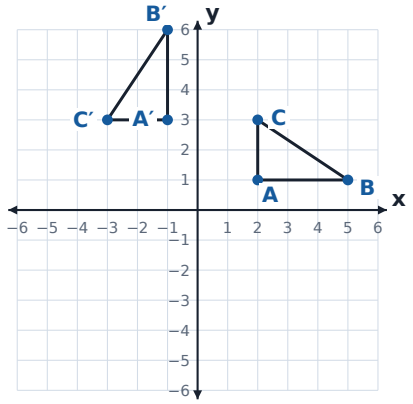
Reflect across the x-axis, then translate 3 units left.

GRADING NOTE Any valid sequence (for example, translate 3 units left, then reflect across the x-axis). The orientation is reversed, so the sequence must include a reflection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

333

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

Rotate 90° counterclockwise about the origin, then translate 1 unit up.

GRADING NOTE Any valid sequence. Check: (x, y) → (−y, x) → (−y, x + 1) sends A(2, 1) to (−1, 3), B to (−1, 6), C to (−3, 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

334

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

GRADING NOTE The rotation sends B to (0, 2) and C to (−3, 0); the translation then gives B′(4, 2) and C′(1, 0). The other orders or turns send B or C elsewhere.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

True or false?

Triangles PQR and P′Q′R′ are congruent. P, Q, R are in clockwise order and P′, Q′, R′ are in counterclockwise order. Some sequence of rotations and translations alone carries P to P′, Q to Q′, and R to R′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

335

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rotations and translations keep the order (orientation) of the vertices; a reflection is needed to reverse it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A translation

C. A rotation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

337

QUESTION ID

S10-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. A translation

GRADING NOTE Sliding every point the same way is a translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

338

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

339

QUESTION ID

S10-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep all lengths and angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

340

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A translation 2 units up

B. A reflection across the x-axis

C. A 90° counterclockwise rotation about the origin

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

341

QUESTION ID

S10-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. A 90° counterclockwise rotation about the origin

GRADING NOTE (2, 0) → (0, 2) under (x, y) → (−y, x), and the origin stays fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

343

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point (−1, 4) is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

344

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−4, −1)

GRADING NOTE Form required: an ordered pair. (x, y) → (−y, x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (3, −2)

B. (7, 4)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

345

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, −2)

GRADING NOTE (5 − 2, 1 − 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

346

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, −y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line y = x is (4, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

347

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (1, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

349

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Counterclockwise

GRADING NOTE Any equivalent wording. From A go right to B, then up-left to C: a counterclockwise turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

350

QUESTION ID

S10-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A mirror image runs the other way around.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

351

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rotations keep the order of the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. counterclockwise

B. either, depending on the lines

C. clockwise

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

352

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. clockwise

GRADING NOTE Each reflection reverses the order, so two reflections restore it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

353

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent wording. The translation keeps the order; the reflection reverses it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

355

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A translation of 6 units, perpendicular to the lines (in the direction from the first line toward the second).

GRADING NOTE Any equivalent wording with the distance (twice the gap) and the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

356

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation of 80° about O (in the direction from the first line toward the second).

GRADING NOTE Any equivalent wording with the center and twice the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

357

QUESTION ID

S10-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Rotations and translations keep orientation, but these triangles have opposite orientations, so a reflection is needed.

GRADING NOTE Any equivalent explanation based on orientation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

358

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a known theorem: a reflection fixes one vertex, a second fixes two, and a third (if needed) fixes the third vertex of a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

359

QUESTION ID

S10-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE The orientation is kept, so an even number is needed; two reflections give any rotation or translation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

361

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Dilate by a factor of 3 about the origin, then rotate 90° counterclockwise about the origin.

GRADING NOTE Any valid sequence. Check: (2, 0) → (6, 0) → (0, 6) and (0, 1) → (0, 3) → (−3, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F2-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

362

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Dilate by 2 about the origin, then translate 2 units right and 2 units up.

GRADING NOTE Any valid sequence. (x, y) → (2x + 2, 2y + 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

363

QUESTION ID
S10-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are similar; they are congruent only when the scale factor is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

364

QUESTION ID

S10-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

GRADING NOTE (3, 0) → (6, 0) → (0, 6) → (1, 7), and (0, 0) → (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

365

QUESTION ID

S10-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. Each side of XYZ is 2 times the matching side of PQR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

368

QUESTION ID

S11-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 7 and 48°

GRADING NOTE A reflection is a rigid motion: it keeps every length and every angle measure, wherever ℓ is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

369

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

P′Q′ ∥ S′R′. A rotation maps lines to lines and keeps them from meeting if they did not meet before, so parallel lines stay parallel.

GRADING NOTE Any equivalent explanation. Essential: the images are parallel because rigid motions take parallel lines to parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

370

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

4 cm, on the other side of m.

GRADING NOTE Any equivalent wording. m is the perpendicular bisector of AA′.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

True or false?

A translation can reverse the orientation (the clockwise or counterclockwise order) of a triangle's vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

371

QUESTION ID
S11-M-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A translation slides every point the same way, so the order of the vertices around the triangle is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

372

QUESTION ID

S11-M-Q5

TYPE

Short answer

CORRECT ANSWER

On ℓ. A 180° rotation about a point of ℓ sends every point of ℓ to the point of ℓ on the opposite side of O, so ℓ maps onto itself.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

374

QUESTION ID

S11-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions map lines to lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

375

QUESTION ID

S11-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep parallel lines parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

376

QUESTION ID

S11-B1-Q3

TYPE

Short answer

CORRECT ANSWER

52°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 30 cm

B. 4.5 cm

C. 39 cm

D. 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

377

QUESTION ID

S11-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9 cm

GRADING NOTE Rotations keep lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

378

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Any two of: side lengths, angle measures, area, parallel sides, perimeter.

GRADING NOTE Any two correct properties.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

380

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

B′C′

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠Q′

D. ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

381

QUESTION ID

S11-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. ∠Q′

GRADING NOTE The prime marks the image of each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

382

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding sides of an image under a rigid motion are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

383

QUESTION ID

S11-B2-Q4

TYPE

Short answer

CORRECT ANSWER

D′E′ = 6, E′F′ = 9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

384

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is usually in a new position; it only has the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

386

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Counterclockwise. Each reflection reverses the orientation, so two reflections restore it.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° about point O and then rotated 50° about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

387

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation of 120° about O (in the same direction).

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity has exactly one fixed point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

388

QUESTION ID

S11-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every point moves by the same nonzero amount, so no point is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A reflection across line ℓ

C. A rotation of 90° about point O

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

389

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A rotation of 90° about point O

GRADING NOTE A 90° rotation fixes only its center; a reflection fixes every point of its line; a translation fixes none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

390

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

ℓ itself. Each point of ℓ goes to the point of ℓ on the other side of m, the same distance away.

GRADING NOTE Any equivalent explanation (a line perpendicular to the mirror maps onto itself).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

392

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

OC = 5 and CD = 7.

GRADING NOTE Any equivalent form. The rotation sends OA to OC and AB to CD, keeping lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

393

QUESTION ID
S11-F2-Q2

TYPE
Short answer

CORRECT ANSWER
64°

GRADING NOTE Any equivalent form. ∠PQR maps to ∠PSR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

394

QUESTION ID

S11-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rigid motions keep lengths, so AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL ∥ K′L′ or KL is on line K′L′

B. ∠L ≅ ∠L′

C. KL is on the same line as K′L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

395

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. KL is on the same line as K′L′

GRADING NOTE A translation keeps length and angle and keeps lines parallel to themselves, but the image is usually on a different line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

396

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The two equal sides map to two sides of the same lengths, so the image has two equal sides.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

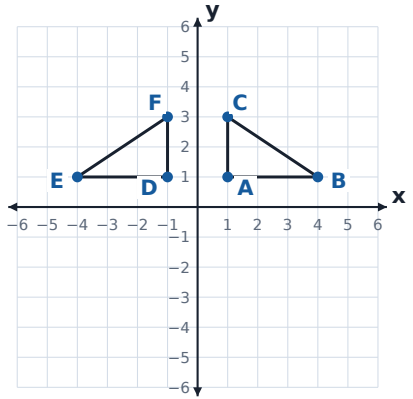
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

399

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

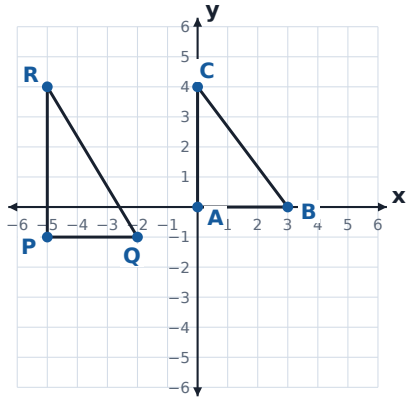
Yes. The reflection across the y-axis carries A to D, B to E, and C to F.

GRADING NOTE Any valid rigid motion or sequence. The conclusion and a motion that works are both required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

400

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. ABC has legs 3 and 4, but PQR has legs 3 and 5. A rigid motion keeps every length, so none can carry one onto the other.

GRADING NOTE Form required: "No" with a length (or area) that differs. Areas 6 and 7.5 also show it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Which fact proves that two figures are NOT congruent?

A. One is above the other

B. One has area 12 and the other has area 15

C. They face opposite directions

D. They are in different quadrants

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

401

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. One has area 12 and the other has area 15

GRADING NOTE Rigid motions keep area. Direction and position can be changed by rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

True or false?

Two figures are congruent if some sequence of rotations, reflections, and translations carries one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

402

QUESTION ID

S12-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of congruence in terms of rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

403

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. A 90° rotation (followed by a translation if needed) carries R onto S.

GRADING NOTE Any valid rigid motion. Both are 4 by 6 rectangles; turning one a quarter turn matches the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

405

QUESTION ID

S12-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A translation is a rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

406

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are similar, not congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A triangle and its reflection

C. A circle of radius 2 and a circle of radius 3

D. A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

407

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A triangle and its reflection

GRADING NOTE A reflection keeps size and shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

408

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5, 6, and 7

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

409

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer: both are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

411

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

412

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from (−2, 1) to (1, 5).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

413

QUESTION ID

S12-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $\sqrt{(3^2 + 4^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

414

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both are √5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

415

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its length is $\sqrt{18} = 3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

417

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Translate P's center to Q's center. Every point 3 units from P's center goes to a point 3 units from Q's center, so circle P lands on circle Q.

GRADING NOTE Any equivalent explanation using a translation that matches the centers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

418

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. A rigid motion keeps angle measures, but one has 90° angles and the other a 60° angle.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

419

QUESTION ID

S12-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A 3-4-5 triangle and a 4-4-4 triangle both have perimeter 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Which information is enough to show two segments are congruent?

A. They have the same length

B. They have the same slope

C. They have the same midpoint

D. They both start at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

420

QUESTION ID

S12-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. They have the same length

GRADING NOTE A rigid motion can carry any segment onto any other segment of the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

421

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Translate the vertex of angle 1 to the vertex of angle 2, rotate so one side lies along a side of angle 2, and reflect across that side if the other side is on the wrong side.

GRADING NOTE Any valid description that matches vertex and both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

423

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

EF = 6, m∠G = 110°, CD = 4.

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

424

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form. RS = XY, so 2x + 3 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

425

QUESTION ID

S12-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ∠B corresponds to ∠E.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. ∠S and ∠S′

B. PQ and RS

C. QR and S′T′

D. ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

426

QUESTION ID
S12-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. ∠S and ∠S′

GRADING NOTE Each part corresponds to its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

427

QUESTION ID

S12-F2-Q5

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE Any equivalent form. ∠BAC corresponds to ∠DCA, so 3y = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries △ABC onto △DEF with A → D, B → E, C → F. Explain why AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

430

QUESTION ID

S13-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

Each side and angle of ABC is carried onto the corresponding side and angle of DEF, and rigid motions keep lengths and angle measures.

GRADING NOTE

REQUIRED WORK: Use the definition: the motion sends each vertex to its partner, so it sends each side and angle onto the corresponding one, and rigid motions keep length and angle measure.

Any valid explanation. Essential: (1) the motion maps segment AB onto segment DE (endpoints to endpoints), and similarly for the other sides and angles; (2) rigid motions preserve distance and angle measure. Invalid: "they look the same".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In △ABC and △DEF, AB = DE, BC = EF, CA = FD, ∠A ≅ ∠D, ∠B ≅ ∠E, and ∠C ≅ ∠F. Explain how to build a rigid motion that carries △ABC onto △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

431

QUESTION ID

S13-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and (if needed) a reflection carry A to D, B to E, and C to F.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate about D so ray AB lies on ray DE (B lands on E because AB = DE); if C is on the other side of line DE from F, reflect across DE. Explain why C then lands on F.

Any valid explanation. Essential: the translation and rotation match A and B; the reflection (if needed) puts C on F's side; C lands on F because ∠A ≅ ∠D puts C on ray DF and AC = DF fixes its distance from D. Accept using ∠B and ∠E instead. Invalid: stating the result without explaining why C lands on F.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

A rigid motion maps △PQR onto △XYZ with P → X, Q → Y, R → Z. Which must be true?

A. QR = YZ

B. PQ = YZ

C. ∠P ≅ ∠Y

D. QR = XZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

QUESTION ID

S13-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. QR = YZ

GRADING NOTE Side QR corresponds to side YZ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

△JKL ≅ △MNO with J ↔ M, K ↔ N, L ↔ O. JK = 8, KL = 11, and m∠L = 37°. Find MN, NO, and m∠O.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

433

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

MN = 8, NO = 11, m∠O = 37°

GRADING NOTE Any equivalent form. Corresponding parts of congruent triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

True or false?

Two triangles whose three pairs of corresponding angles are congruent must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

434

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A small and a large equilateral triangle have the same angles but are not congruent: the sides must match too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

436

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠Y

GRADING NOTE Any equivalent form. The order of the letters gives the correspondence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

437

QUESTION ID

S13-B1-Q2

TYPE

Short answer

CORRECT ANSWER

XZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and MN

B. QR and LN

C. PQ and MN

D. PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

438

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. QR and MN

GRADING NOTE Q ↔ M and R ↔ N.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

△DEF ≅ △GHI means that ∠F ≅ ∠I.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

439

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The third letters correspond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

△DEF ≅ △GHI means that DE = HI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

440

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE DE corresponds to GH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

442

QUESTION ID

S13-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotations are rigid motions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

443

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 units

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

444

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 71°

GRADING NOTE Reflections keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

445

GRADING NOTE Translations keep every length.

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

446

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Translations, rotations, and reflections

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

448

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 units right and 4 units down

GRADING NOTE Any equivalent form: (x, y) → (x + 3, y − 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

449

QUESTION ID

S13-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90° counterclockwise

GRADING NOTE Any equivalent form (270° clockwise).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A translation along DE

C. A rotation about the midpoint of DE

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

450

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A reflection across line DE

GRADING NOTE A reflection across DE keeps every point of the line fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

451

QUESTION ID

S13-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Points on the mirror line are their own images.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

452

QUESTION ID

S13-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A nonzero translation moves every point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

454

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Three congruent sides force the angles to be congruent too (SSS), so all six parts match and a rigid motion exists.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

455

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Triangles with sides 6 and 8 and a 30° angle opposite the 6-unit side: the 8-unit side can make the third side either long or short (SSA).

GRADING NOTE Any valid SSA counterexample with the angle not between the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

456

QUESTION ID

S13-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE AAA gives similar triangles, not necessarily congruent ones.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Three sides

B. Two sides and an angle not between them

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

457

QUESTION ID

S13-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Two sides and an angle not between them

GRADING NOTE SSA can give two different triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

458

QUESTION ID

S13-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. The angle sum is 180° in both triangles, so ∠C = 180° − ∠A − ∠B = ∠F.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

460

QUESTION ID

S13-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠ABD ≅ ∠CBD, so BD bisects ∠ABC.

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △CBD (SSS, with BD shared), then use corresponding angles.

Any valid proof. Essential: AB = CB, AD = CD, BD = BD (reflexive); SSS gives the congruence; corresponding angles ∠ABD and ∠CBD are congruent, which is the definition of a bisector. Invalid: assuming the angles are equal from the picture.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

461

QUESTION ID

S13-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = CD

GRADING NOTE REQUIRED WORK: Use vertical angles and the bisected segments to prove △AEB ≅ △CED by SAS.

Any valid proof. Essential: AE = CE and BE = DE (definition of bisect); ∠AEB ≅ ∠CED (vertical angles); SAS; corresponding sides AB = CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In △PQR and △STU, PQ = ST, QR = TU, and PR = SU. The triangles are congruent. What must be true of ∠R and ∠U? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

462

QUESTION ID
S13-F2-Q3

TYPE
Short answer

CORRECT ANSWER
∠R ≅ ∠U, because corresponding angles of congruent triangles are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

463

QUESTION ID
S13-F2-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Congruence gives all six pairs of corresponding parts (CPCTC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XM and YM

D. XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

464

QUESTION ID

S13-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. XM and YM

GRADING NOTE Midpoint means XM = MY.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

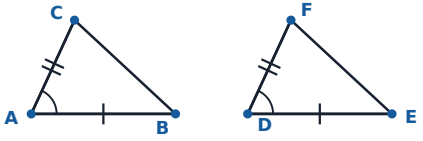
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, AC = DF, and ∠A ≅ ∠D. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

467

QUESTION ID

S14-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F, so the triangles are congruent.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE; reflect across DE if C is on the other side from F. Then explain why B lands on E and C lands on F.

Any valid explanation. Essential: (1) translate A onto D; (2) rotate about D so ray AB lies on ray DE; B lands on E because AB = DE; (3) reflect across DE if needed so C is on F's side; (4) ray AC lands on ray DF because ∠A ≅ ∠D; (5) C lands on F because AC = DF. Invalid: "the triangles look the same"; citing SAS to prove SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In the rigid-motion proof of SAS, A has been moved to D and ray AB lies on ray DE, with C on the same side of line DE as F. Why must ray AC lie on ray DF?

A. BC = EF

B. The triangles have equal areas

C. AC = DF

D. ∠A ≅ ∠D, so both rays make the same angle with ray DE on the same side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

468

QUESTION ID

S14-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. ∠A ≅ ∠D, so both rays make the same angle with ray DE on the same side

GRADING NOTE There is only one ray from D on that side of line DE making that angle with ray DE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

In the rigid-motion proof of SAS, ray AC now lies on ray DF. Why does C land exactly on F?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

469

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

AC = DF and rigid motions keep length, so C is the point of ray DF at distance DF from D, which is F.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

Explain why a reflection is sometimes needed in the rigid-motion proof of SAS, and why it does not move D or E.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

470

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

After the rotation, C may be on the opposite side of line DE from F. Reflecting across line DE moves it to F's side; D and E are on the line, so they stay fixed.

GRADING NOTE Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

True or false?

The rigid-motion argument for SAS still works if the congruent angle is not between the two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

471

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Then the angle does not fix where the third vertex goes; SSA can give two different triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

473

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠A

GRADING NOTE Any equivalent form. The included angle is at the shared vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

474

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

PQ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

475

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ∠X

GRADING NOTE Sides XY and YZ share vertex Y, so ∠Y is included; ∠X is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

476

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two sides and the included angle fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

477

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE When the angle is not between the sides, there may be two triangles or none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

479

QUESTION ID

S14-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions map rays to rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

480

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. 5 units from D on the image ray

D. farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

481

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units from D on the image ray

GRADING NOTE A rotation about D keeps distances from D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

482

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the mirror line stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

483

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

485

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A translation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

486

QUESTION ID

S14-B3-Q2

TYPE

Short answer

CORRECT ANSWER

A rotation about D

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A rotation of 180° about E

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

487

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A reflection across line DE

GRADING NOTE Only the reflection keeps both D and E fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

488

QUESTION ID

S14-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The center of a rotation is its own image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

489

QUESTION ID

S14-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It moves A onto D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

491

QUESTION ID

S14-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △CDA by SAS.

GRADING NOTE

REQUIRED WORK: Use alternate interior angles with the parallel sides and the shared diagonal to apply SAS.

Any valid proof. Essential: AB = CD (given); ∠BAC ≅ ∠DCA (alternate interior angles, AB ∥ DC); AC = CA (reflexive); SAS with the angle between the sides. Invalid: using ∠B ≅ ∠D without proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ. Prove that $\triangle PMA \cong \triangle PMB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

492

QUESTION ID

S14-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

$\triangle PMA \cong \triangle PMB$ by SAS, so PA = PB.

GRADING NOTE REQUIRED WORK: Use the midpoint, the right angles, and the shared side PM with SAS.

Any valid proof. Essential: AM = BM (midpoint); $\angle PMA \cong \angle PMB$ (both right angles); PM = PM; SAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

493

QUESTION ID

S14-F1-Q3

TYPE

Short answer

CORRECT ANSWER

△AEB ≅ △CED by SAS (the vertical angles at E are the included angles).

GRADING NOTE Any equivalent correct pair (also △AED ≅ △CEB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

494

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The shared side is congruent to itself (reflexive property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. The angles are both right angles

B. Vertical angles are congruent

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

495

QUESTION ID

S14-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent

GRADING NOTE ∠PTR and ∠QTS (or the other pair) are vertical angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

497

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

∠B ≅ ∠E

GRADING NOTE Form required: the included angle pair. (AC = DF would give SSS, not SAS.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △PQR and △XYZ, PQ = XY, PR = XZ, and ∠Q ≅ ∠Y. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

498

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The congruent angle is not between the congruent sides (SSA), which does not guarantee congruence.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In △ABC and △ADC (sharing AC), AB = AD and ∠BCA ≅ ∠DCA. These facts prove △ABC ≅ △ADC by SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

499

QUESTION ID
S14-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The congruent angles at C are not included between AB and AC (this is SSA), so SAS does not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABC and △EFG

C. △ABD and △EFH

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

500

QUESTION ID

S14-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. △ABC and △EFG

GRADING NOTE The given parts are the two sides and the included angle of △ABC and △EFG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

501

QUESTION ID

S14-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No. SAS guarantees congruence; the third side is then determined (and equal).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

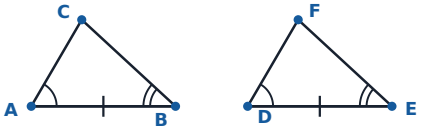
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, ∠A ≅ ∠D, AB = DE, and ∠B ≅ ∠E. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

504

QUESTION ID

S15-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation, a rotation, and possibly a reflection carry A to D, B to E, and C to F.

GRADING NOTE REQUIRED WORK: Translate A to D; rotate so ray AB lies on ray DE (B lands on E); reflect across DE if needed; then use the two angles to place rays AC and BC on rays DF and EF.

Any valid explanation. Essential: (1) A → D and B → E using AB = DE; (2) C on F's side of DE (reflect if needed); (3) ray AC lands on ray DF because ∠A ≅ ∠D; (4) ray BC lands on ray EF because ∠B ≅ ∠E; (5) C is where rays AC and BC meet, so it lands where DF and EF meet, which is F. Invalid: using side lengths AC or BC (not given); citing ASA to prove ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on F's side of line DE. Why does C land on F?

A. C and F are both the highest points

B. BC = EF

C. C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F

D. AC = DF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

505

QUESTION ID
S15-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F

GRADING NOTE The two angles place the two rays; the intersection point is unique.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on the same side of line DE as F. Explain why ray BC lies on ray EF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

506

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

∠B ≅ ∠E, and on that side of line DE there is only one ray from E making that angle with ray ED, namely ray EF.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In the rigid-motion proof of ASA, why can rays DF and EF meet in only one point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

507

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

They lie on two different lines, and two different lines meet in at most one point.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

True or false?

In the rigid-motion proof of ASA, a reflection across line DE leaves D and E where they are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

508

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Points on the line of reflection are fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

510

QUESTION ID

S15-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The side and the two angles at its ends fix where the other two sides meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

511

QUESTION ID

S15-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two angles must add to less than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

512

QUESTION ID

S15-B1-Q3

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. Angles 30° and 80° with a 4 cm side between them

B. One angle of 45° and nothing else

C. Angles 30°, 60°, and 90°

D. Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

513

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Angles 30° and 80° with a 4 cm side between them

GRADING NOTE Two angles with the included side fix the triangle; three angles alone fix only its shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

514

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No. 95° + 90° = 185° is more than 180°.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines y = x + 1 and y = −x + 3 have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

516

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. They meet at (1, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

517

QUESTION ID

S15-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two different lines meet in at most one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines y = 2x + 1 and y = 2x − 4

A. never meet

B. are the same line

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

518

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. never meet

GRADING NOTE Same slope, different intercepts: parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

519

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two points determine one line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where do the lines x = 2 and y = −3 meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

520

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(2, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

522

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

64°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

523

QUESTION ID

S15-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

524

QUESTION ID

S15-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Exactly one

GRADING NOTE On one side of a line there is one ray at a given angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

525

QUESTION ID

S15-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Angle measure does not depend on side length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

526

QUESTION ID

S15-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4 cm from D (on the image ray).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC and △DEF, ∠A ≅ ∠D, ∠B ≅ ∠E, and BC = EF (the side is NOT between the angles). Prove △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

528

QUESTION ID

S15-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △DEF by ASA after showing ∠C ≅ ∠F.

GRADING NOTE REQUIRED WORK: Use the triangle angle sum to show ∠C ≅ ∠F, then apply ASA with ∠B, BC, ∠C.

Any valid proof. Essential: m∠C = 180° − m∠A − m∠B = 180° − m∠D − m∠E = m∠F; then ∠B, side BC, ∠C match ∠E, EF, ∠F (ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

529

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, by AAS: the third angles are both 65°, so ASA applies.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

True or false?

AAS is a valid congruence criterion only because the angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

530

QUESTION ID

S15-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The angle sum gives the third pair of angles, turning AAS into ASA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In △PQR and △XYZ, ∠P ≅ ∠X, ∠Q ≅ ∠Y, and PR = XZ. Which criterion proves congruence?

A. ASA

B. AAS

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

531

QUESTION ID

S15-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. AAS

GRADING NOTE PR is not between ∠P and ∠Q, so it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if ∠A ≅ ∠D and ∠C ≅ ∠F?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

532

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

AB = DE or BC = EF

GRADING NOTE Any side not between the two angles (AC = DF would be ASA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

534

QUESTION ID

S15-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABE ≅ △DCE by ASA.

GRADING NOTE REQUIRED WORK: Use alternate interior angles, the vertical angles at E, and AE = DE with ASA.

Any valid proof. Essential: ∠BAE ≅ ∠CDE (alternate interior, AB ∥ CD); AE = DE (midpoint); ∠AEB ≅ ∠DEC (vertical); ASA. AAS with ∠B ≅ ∠C also accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In △ABC, the bisector of ∠A is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

535

QUESTION ID

S15-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = AC

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △ACD by ASA using the bisected angle, the shared side AD, and the right angles.

Any valid proof. Essential: ∠BAD ≅ ∠CAD (bisector); AD = AD; ∠ADB ≅ ∠ADC (right angles); ASA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, ∠A is shared, AB = AC, and ∠B ≅ ∠C. Which criterion proves △ABE ≅ △ACD?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

536

QUESTION ID

S15-F2-Q3

TYPE

Short answer

CORRECT ANSWER

ASA (∠A, AB = AC, ∠B ≅ ∠C)

GRADING NOTE Form required: the criterion with the matching parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

537

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE By the reflexive property, the shared angle is congruent to itself and can be used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

A. JL = JL

B. JK = JM

C. KL = ML

D. JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

538

QUESTION ID

S15-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. JL = JL

GRADING NOTE JL is between the two pairs of angles and is shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

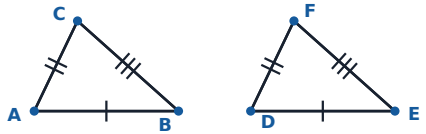
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △ABC and △DEF, AB = DE, BC = EF, and CA = FD. Explain, using rigid motions, why △ABC ≅ △DEF.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

541

QUESTION ID

S16-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

A translation and a rotation carry A to D and B to E; if needed, the reflection across DE carries C′ to F.

GRADING NOTE REQUIRED WORK: Move A to D and B to E (translation and rotation, using AB = DE). If the image C′ of C is not F, show that line DE is the perpendicular bisector of C′F, so the reflection across DE carries C′ to F.

Any valid explanation. Essential: (1) A → D, B → E (AB = DE); (2) DC′ = DF and EC′ = EF (rigid motions keep lengths); (3) if C′ ≠ F, D and E are each equidistant from C′ and F, so line DE is the perpendicular bisector of C′F; (4) the reflection across DE fixes D and E and maps C′ to F. Alternative: C′ and F both lie on the circles centered at D and E with these radii, which meet in at most one point on each side of DE. Invalid: using angles (none are given); citing SSS to prove SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In the rigid-motion proof of SSS, A is on D, B is on E, and C′ (the image of C) is not F. Why is line DE the perpendicular bisector of C′F?

A. DE bisects ∠D

B. DE is parallel to C′F

C. D and E are each the same distance from C′ as from F

D. C′F is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

542

QUESTION ID

S16-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. D and E are each the same distance from C′ as from F

GRADING NOTE A point equidistant from C′ and F lies on the perpendicular bisector of C′F; two such points determine that line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

In the rigid-motion proof of SSS, line DE is the perpendicular bisector of C′F. Why does the reflection across DE carry C′ onto F, and why is that enough?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

543

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

By the definition of a reflection, it maps C′ to F because DE is the perpendicular bisector of C′F; it keeps D and E fixed, so the whole triangle lands on DEF.

GRADING NOTE Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

The rigid-motion proof of SSS uses no angle information. Explain how the three side lengths alone locate the third vertex.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

544

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

Once D and E are matched, the third vertex must be DF from D and EF from E: it lies on two circles, which meet in at most one point on each side of DE.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

True or false?

A triangle with sides 5, 7, and 9 is congruent to a triangle with sides 9, 5, and 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

545

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Matching the sides by length gives three pairs of congruent sides (SSS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

547

QUESTION ID

S16-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 + 4 = 7 is less than 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

548

QUESTION ID

S16-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 + 6 > 10, and three sides fix the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

549

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4, 5, 7

GRADING NOTE Each pair must add to more than the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

550

QUESTION ID

S16-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Between 4 and 16

(4 < x < 16)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

551

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No. Three side lengths determine one triangle shape and size.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

553

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. The circles of radius 5 about A and B meet twice, once on each side of AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

All points 3 cm from a point A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

554

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is the definition of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. Both on line AB

B. One on each side of line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

555

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. One on each side of line AB

GRADING NOTE The picture is symmetric across line AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

556

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They can meet in 0, 1, or 2 points (or be the same circle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

557

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

No. 3 + 4 = 7 < 10, so the circles do not meet.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

559

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 5, and 7

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

560

QUESTION ID

S16-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rigid motions keep all distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

A. The perimeter

B. The length of a side

C. The area

D. The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

561

QUESTION ID

S16-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The position of a triangle

GRADING NOTE Only the position changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

562

QUESTION ID

S16-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The image is the same distance from the line, on the other side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

563

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6 units

GRADING NOTE Any equivalent form. D is on the line, so it is fixed, and distances are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

565

QUESTION ID

S16-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

△ABC ≅ △DEF by SSS (BC = EF from the Pythagorean Theorem).

GRADING NOTE REQUIRED WORK: Use the Pythagorean Theorem to show BC = EF, then apply SSS.

Any valid proof. Essential: BC = √(AB² − AC²) = √(DE² − DF²) = EF; then SSS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

566

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes; the other leg is 12 in both.

GRADING NOTE Form required: the conclusion and the leg. $\sqrt{(169 - 25)} = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F1-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

567

QUESTION ID

S16-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The right angle is opposite the hypotenuse; the Pythagorean Theorem fixes the third side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their right angles are equal

C. Their hypotenuses are equal and one pair of legs is equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

568

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Their hypotenuses are equal and one pair of legs is equal

GRADING NOTE HL needs the hypotenuse and a leg.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

569

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

PR = XZ = 8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

571

QUESTION ID
S16-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Equilateral triangles with sides 2 and 5 have all angles 60° but are not congruent.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

572

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

With sides 6 and 8 and a 30° angle opposite the 6, the third vertex can be in two places, giving two different triangles.

GRADING NOTE Any valid SSA counterexample (two non-congruent triangles with the same SSA data).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SAS, ASA, AAS

B. SSS, SSA, ASA, AAS

C. SSA, AAA, SAS, HL

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

573

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. SSS, SAS, ASA, AAS

GRADING NOTE SSA and AAA are not criteria (HL is a special right-triangle case).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

574

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE SSS gives congruence, and corresponding angles of congruent triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal areas must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

575

QUESTION ID
S16-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The included angles could be θ and 180° − θ: equal areas (same sine), different third sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

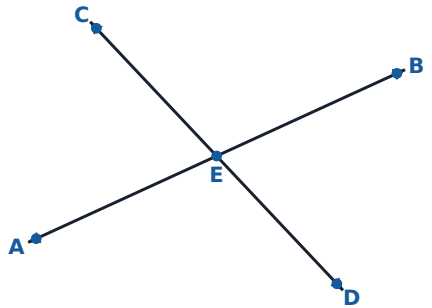
SET 17 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

578

QUESTION ID

S17-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠AEC ≅ ∠BED

GRADING NOTE REQUIRED WORK: Use the two linear pairs that share ∠CEB, and the subtraction property of equality.

Any valid proof (paragraph, two-column, or flow). Essential: (1) m∠AEC + m∠CEB = 180° (linear pair: A, E, B are collinear); (2) m∠CEB + m∠BED = 180° (linear pair: C, E, D are collinear); (3) so m∠AEC + m∠CEB = m∠CEB + m∠BED (substitution/transitive); (4) subtract m∠CEB. Alternative: the 180° rotation about E maps ray EA to ray EB and ray EC to ray ED. Invalid: "they look equal", measuring, or citing the theorem itself.

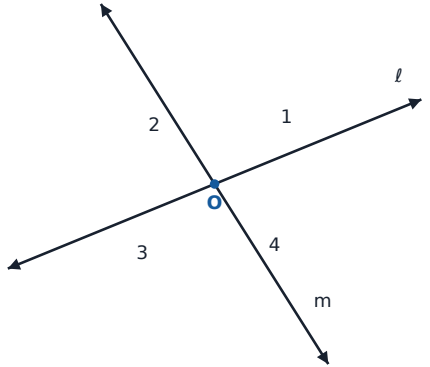
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

579

QUESTION ID

S17-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$\angle 2 \cong \angle 4$

GRADING NOTE REQUIRED WORK: Use the linear pairs $\angle 1$ and $\angle 2$, and $\angle 1$ and $\angle 4$, then the subtraction property of equality.

Any valid proof (paragraph, two-column, or flow). Essential: $m\angle 1 + m\angle 2 = 180°$ and $m\angle 1 + m\angle 4 = 180°$ (linear pairs, each with a reason); so $m\angle 2 = 180° - m\angle 1 = m\angle 4$. Using $\angle 3$ as the shared angle is also valid. Invalid: using $\angle 2 \cong \angle 4$ as a step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A proof that ∠1 ≅ ∠3 for vertical angles 1 and 3 (with ∠2 next to both):

1. m∠1 + m∠2 = 180° (linear pair)
2. m∠2 + m∠3 = 180° (linear pair)
3. m∠1 + m∠2 = m∠2 + m∠3 (?)
4. m∠1 = m∠3 (?)

Give the reasons for steps 3 and 4.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

580

QUESTION ID
S17-M-Q3

TYPE
Short answer

CORRECT ANSWER
Step 3: substitution (or the transitive property of equality); step 4: subtraction property of equality.

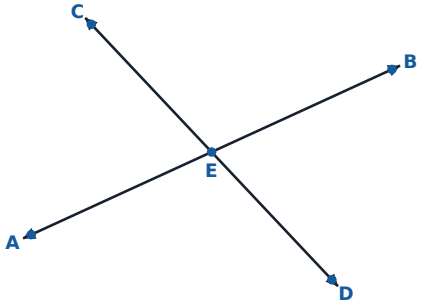
GRADING NOTE Any equivalent names for both reasons. Both are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

581

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The 180° rotation about E maps ∠AEC onto ∠BED, so the angles are congruent.

GRADING NOTE

REQUIRED WORK: Use the 180° rotation about E and explain where it sends rays EA and EC.

Any valid explanation. Essential: a 180° rotation about E maps each line through E onto itself, sending ray EA to the opposite ray EB and ray EC to ray ED; so it maps ∠AEC onto ∠BED; rotations keep angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

A student measures two vertical angles in one drawing, finds that both are 47°, and writes "vertical angles are congruent." Why is this not a proof?

A. Protractors are never accurate

B. Vertical angles are not always congruent

C. It checks one example; a proof must show the result for every pair of vertical angles

D. The student should have measured all four angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

582

QUESTION ID
S17-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. It checks one example; a proof must show the result for every pair of vertical angles

GRADING NOTE Measuring can suggest a theorem but cannot prove it for all cases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

584

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

52°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

585

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form. 4x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

586

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Together they form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠BAD and ∠BCD

C. ∠ABD and ∠DBC

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

587

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ∠ABD and ∠DBC

GRADING NOTE They share ray BD and their other sides are opposite rays BA and BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

588

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Supplementary angles need not be adjacent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From m∠1 + 40° = m∠2 + 40°, which property gives m∠1 = m∠2?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

590

QUESTION ID
S17-B2-Q1

TYPE
Short answer

CORRECT ANSWER
Subtraction property of equality

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q1**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If x + y = 180 and y + z = 180, which property lets you write x + y = y + z?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

591

QUESTION ID
S17-B2-Q2

TYPE
Short answer

CORRECT ANSWER
Substitution (or the transitive property of equality)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-B2-Q2**

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Symmetric property

C. Distributive property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

592

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Symmetric property

GRADING NOTE Swapping the sides is the symmetric property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as m∠A = m∠A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

593

QUESTION ID
S17-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE a = a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From 2x = 2y you may conclude x = y by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

594

QUESTION ID
S17-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE That step divides by 2 (division property).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q5

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

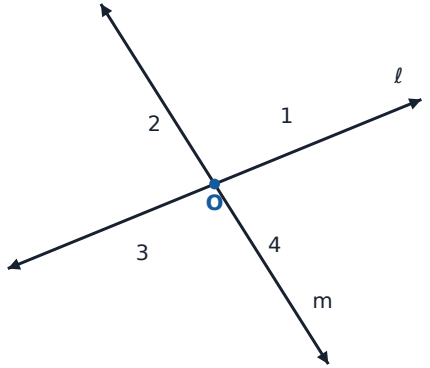
BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines ℓ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Which angle is vertical to ∠1?

A. ∠3 **B.** ∠1

C. ∠4 **D.** ∠2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

596

QUESTION ID

S17-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

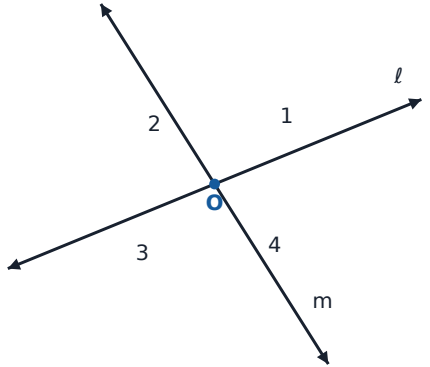
A. ∠3

GRADING NOTE Vertical angles are opposite each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines ℓ and m meet at O, forming ∠1, ∠2, ∠3, ∠4 in order. Name the two angles that form linear pairs with ∠2.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

597

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

∠1 and ∠3

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

598

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They share only the vertex; their sides are opposite rays.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

599

QUESTION ID

S17-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ∠1 with ∠3, and ∠2 with ∠4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

600

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Vertical: 63°; adjacent: 117° and 117°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

602

QUESTION ID

S17-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠BEG = m∠GED, so EG bisects ∠BED.

GRADING NOTE REQUIRED WORK: Combine the bisector definition with vertical angles (∠AEF and ∠BEG; ∠FEC and ∠GED).

Any valid proof (paragraph, two-column, or flow). Essential: m∠AEF = m∠FEC (bisector); ∠AEF ≅ ∠BEG and ∠FEC ≅ ∠DEG (vertical angles, since EG is opposite EF, EB opposite EA, ED opposite EC); so m∠BEG = m∠GED.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

603

QUESTION ID

S17-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

The angles in order are 50°, 70°, 60°, 50°, 70°, 60°.

GRADING NOTE REQUIRED WORK: Use the straight angle (180°) to find the third angle on one side, then vertical angles for the other three.

Any valid proof (paragraph, two-column, or flow). Essential: three adjacent angles on one side of a line add to 180°, so the third is 60°; each of the other three is vertical to one of these.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

604

QUESTION ID

S17-F1-Q3

TYPE

Short answer

CORRECT ANSWER

x = 20; angles 90°, 90°, 90°, 90°

GRADING NOTE Form required: x and the four angles. 4x + 10 = 6x − 30 gives x = 20, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

605

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its vertical angle is 90° and its neighbors are 180° − 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 72°

B. 144°

C. 36°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

606

QUESTION ID

S17-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 72°

GRADING NOTE 2x + 3x = 180, so x = 36 and m∠AEC = 72°; ∠BED is vertical to ∠AEC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

608

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠EBF = 90°

GRADING NOTE REQUIRED WORK: Express m∠EBF as half of m∠ABD plus half of m∠DBC.

Any valid proof (paragraph, two-column, or flow). Essential: m∠EBD = ½m∠ABD, m∠DBF = ½m∠DBC (bisectors); m∠EBF = ½(m∠ABD + m∠DBC) = ½(180°) = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

609

QUESTION ID
S17-F2-Q2

TYPE
Short answer

CORRECT ANSWER
No. Two 30° angles in different places in a figure are congruent but not vertical angles.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

610

QUESTION ID

S17-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

The bisector rays are opposite rays, so they lie on one line.

GRADING NOTE REQUIRED WORK: Show that the angle between the two bisector rays is 180°.

Any valid proof (paragraph, two-column, or flow). Essential: if the vertical angles each measure 2a and the other two each 180° − 2a, the angle from one bisector to the other is a + (180° − 2a) + a = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of a pair of vertical angles are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

611

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They form one straight line: each bisector is the extension of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. It depends on the lines

C. 180°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

612

QUESTION ID

S17-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 90°

GRADING NOTE Adjacent angles formed by two lines are a linear pair, so their bisectors meet at 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

Prove the angle theorems for parallel lines cut by a transversal

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

SET 18 • HSG-CO.C.9

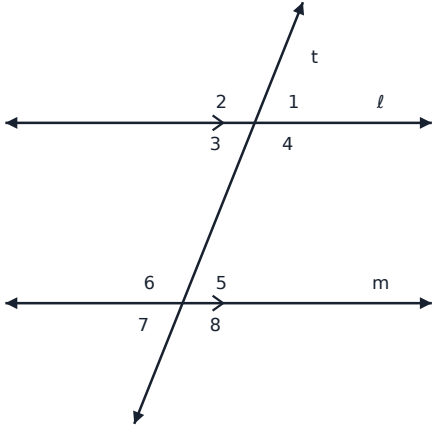
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are parallel and are cut by transversal t. Assume the corresponding angles postulate (corresponding angles formed by a transversal of parallel lines are congruent). Prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

615

QUESTION ID

S18-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

$\angle 3 \cong \angle 5$

GRADING NOTE REQUIRED WORK: Use the corresponding angle of $\angle 5$ at the top crossing and vertical angles.

Any valid proof (paragraph, two-column, or flow). Essential: $\angle 1 \cong \angle 5$ (corresponding angles postulate, $\ell \parallel$ m); $\angle 1 \cong \angle 3$ (vertical angles); so $\angle 3 \cong \angle 5$ (transitive). Invalid: "they look equal"; assuming alternate interior angles are congruent.

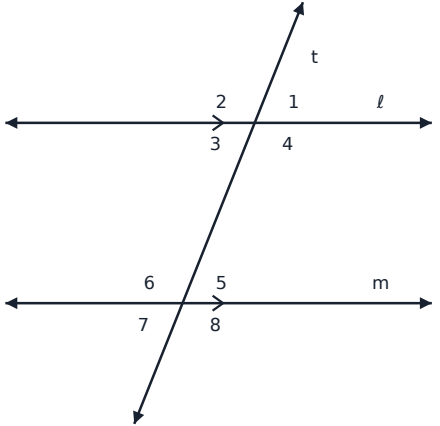
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are parallel and are cut by transversal t. Assume that alternate interior angles are congruent. Prove that corresponding angles $\angle 2$ and $\angle 6$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

616

QUESTION ID

S18-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

$\angle 2 \cong \angle 6$

GRADING NOTE REQUIRED WORK: Use an alternate interior angle and vertical angles.

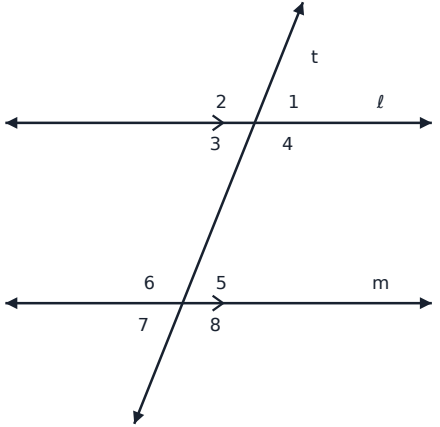
Any valid proof (paragraph, two-column, or flow). Essential: $\angle 4 \cong \angle 6$ (alternate interior angles); $\angle 2 \cong \angle 4$ (vertical angles); so $\angle 2 \cong \angle 6$ (transitive).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

617

QUESTION ID

S18-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The translation that moves Q to P maps m onto ℓ and maps t onto itself, so it maps $\angle 5$ onto $\angle 1$; translations keep angle measure.

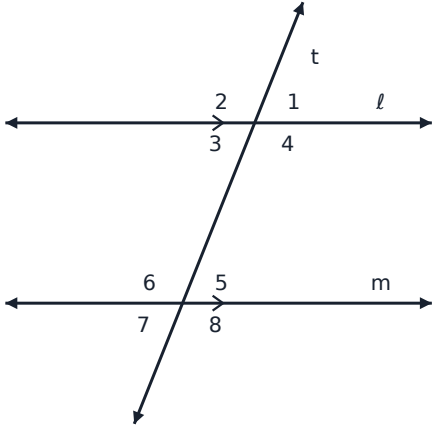
GRADING NOTE REQUIRED WORK: Translate along the transversal from Q to P. Explain why the image of m is ℓ and why the image of $\angle 5$ is $\angle 1$.

Any valid explanation. Essential: (1) the translation maps t onto itself (it moves along t); (2) it maps m to a line through P parallel to m; (3) by the parallel postulate that line is ℓ; (4) so the angle formed by t and m at Q maps to the corresponding angle at P; (5) angle measure is kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

Lines $\ell \parallel m$ are cut by transversal t, and $m\angle 1 = 115°$. Find $m\angle 5$, $m\angle 3$, and $m\angle 6$. Give a reason for each.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

618

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

$m\angle 5 = 115°$ (corresponding to $\angle 1$); $m\angle 3 = 115°$ (vertical to $\angle 1$); $m\angle 6 = 65°$ (linear pair with $\angle 5$).

GRADING NOTE Form required: each measure with a valid reason. Other valid reason chains are accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

A student "proves" that alternate interior angles are congruent by saying: "The lines are parallel because the alternate interior angles are congruent." What is wrong?

A. Nothing; the argument is correct

B. The argument is circular: it assumes the conclusion

C. Alternate interior angles are supplementary

D. The lines must be perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

619

QUESTION ID

S18-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The argument is circular: it assumes the conclusion

GRADING NOTE The statement uses the result it is trying to prove (and states a converse instead).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

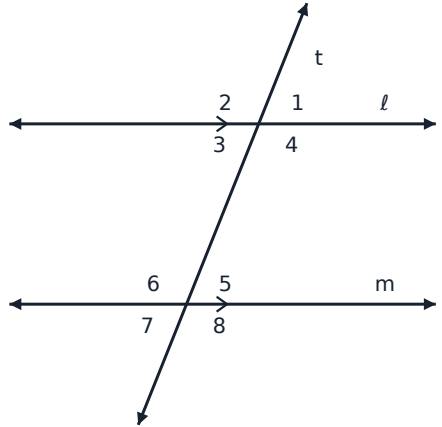
SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

621

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

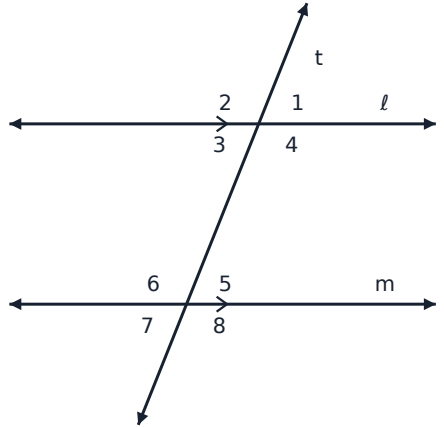
70°

GRADING NOTE Any equivalent form. $\angle 4$ and $\angle 6$ are alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?



A. $\angle 7$

B. $\angle 8$

C. $\angle 6$

D. $\angle 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

622

QUESTION ID

S18-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $\angle 6$

GRADING NOTE Both are in the upper-left position at their crossings.

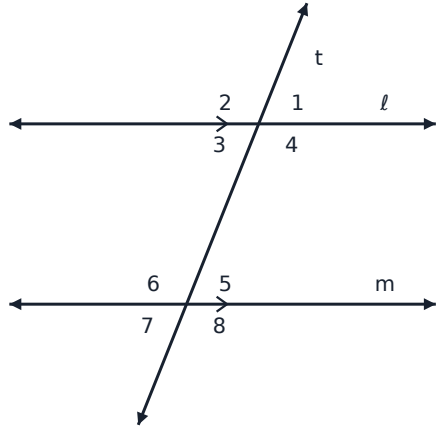
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.

A. True **B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

623

QUESTION ID
S18-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are on the same side of t, between the lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

624

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

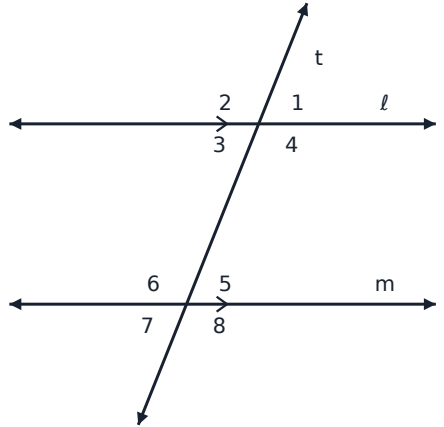
B. False

GRADING NOTE They are congruent only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures $112°$. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

625

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

112°

GRADING NOTE Any equivalent form. $\angle 8$ and $\angle 2$ are alternate exterior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

627

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Vertical angles are congruent

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

628

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

They form a linear pair (supplementary)

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

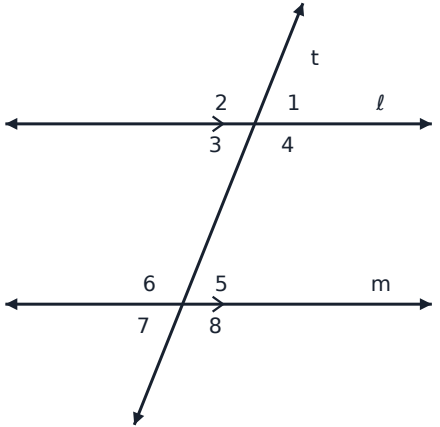
At the top crossing, ∠1 = 110°. What is ∠2 and why?

A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

629

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 70°, linear pair

GRADING NOTE ∠1 and ∠2 are next to each other on line ℓ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

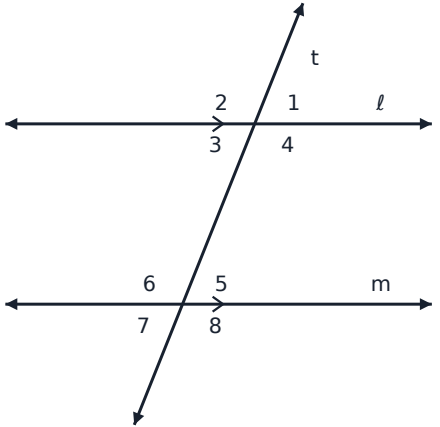
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Angles 1 and 3 are vertical angles in the figure.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

630

QUESTION ID
S18-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are opposite each other at the top crossing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

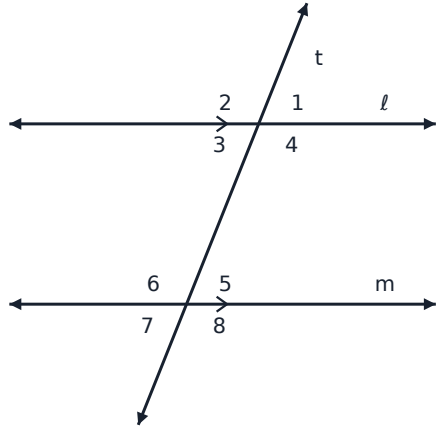
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

631

QUESTION ID
S18-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE They are adjacent: a linear pair along t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

633

QUESTION ID

S18-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

634

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

n is the same line as ℓ.

GRADING NOTE Any equivalent answer: by the parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. x = 2 and y = 2

B. y = x and y = −x

C. y = 2x + 1 and y = −2x + 1

D. y = 2x + 1 and y = 2x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

635

QUESTION ID

S18-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1 and y = 2x − 5

GRADING NOTE Same slope, different intercepts: they never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

636

QUESTION ID

S18-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

637

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Parallel to m (a translation maps a line to a parallel line or itself); so it is ℓ.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSG-CO.C.9

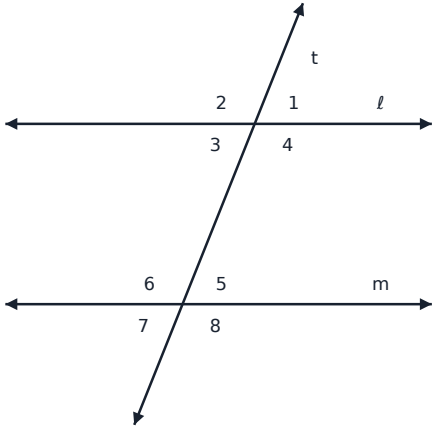
FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

639

QUESTION ID

S18-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

$\ell \parallel$ m

GRADING NOTE

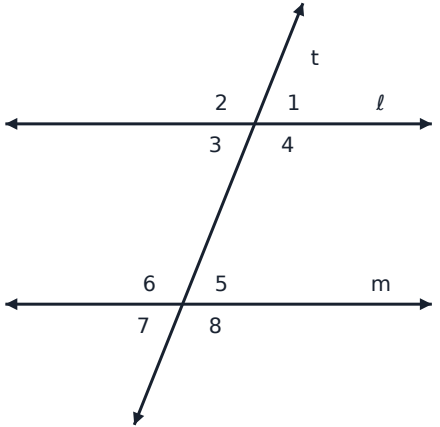
REQUIRED WORK: Indirect proof: suppose ℓ and m meet; they would form a triangle with t whose angles contradict a known result. Or use the 180° rotation about the midpoint of PQ.

Any valid proof (paragraph, two-column, or flow). Essential (indirect): if ℓ and m met at a point R, triangle PQR would have an exterior angle equal to a remote interior angle, which is impossible (an exterior angle is greater than each remote interior angle); so they do not meet. Alternative: the 180° rotation about the midpoint of PQ swaps P and Q and maps ℓ onto m, so if they met on one side they would also meet on the other, giving two common points. Invalid: assuming $\ell \parallel$ m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel$ m? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

640

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 25, because then the alternate interior angles are congruent (105°).

GRADING NOTE Form required: x with the converse as the reason. $5x - 20 = 3x + 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

641

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then the alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of corresponding angles is congruent

B. A pair of vertical angles is congruent

C. A linear pair adds to 180°

D. Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

642

QUESTION ID

S18-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

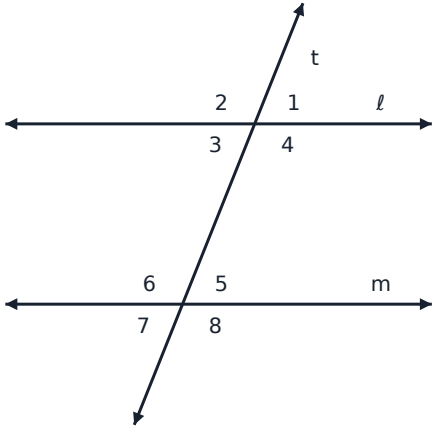
A. A pair of corresponding angles is congruent

GRADING NOTE Vertical angles and linear pairs hold for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

643

QUESTION ID
S18-F1-Q5

TYPE
Short answer

CORRECT ANSWER

No. ∠1 and ∠7 are alternate exterior angles; if $\ell \parallel$ m they would be congruent, but 72° ≠ 108°.

GRADING NOTE Form required: the conclusion with a reason. Equivalently, ∠5 is vertical to ∠7, so $m\angle 5 = 108°$, but its corresponding angle ∠1 is 72°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

645

QUESTION ID

S18-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: Name the two new angles at B, use alternate interior angles, and the straight angle at B.

Any valid proof (paragraph, two-column, or flow). Essential: the line through B parallel to AC exists (parallel postulate); the angles it makes with BA and BC are congruent to ∠A and ∠C (alternate interior angles); those two angles and ∠ABC form a straight angle, 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

646

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

∠A = 50°, ∠C = 65°, ∠B = 65°

GRADING NOTE Form required: all three. Alternate interior angles give ∠A and ∠C; 180 − 50 − 65 = 65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum works without any fact about parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

647

QUESTION ID

S18-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It uses the parallel through one vertex and the alternate interior angle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Alternate interior angles theorem

B. Isosceles triangle theorem

C. Triangle inequality

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

QUESTION ID

S18-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles theorem

GRADING NOTE Sides BA and BC are transversals of the parallel lines (AC and the line through B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

649

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

The exterior angle forms a linear pair with ∠C, so it is 180° − m∠C, which equals m∠A + m∠B by the angle sum.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

SET 19 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

SET 19 • HSG-CO.C.9

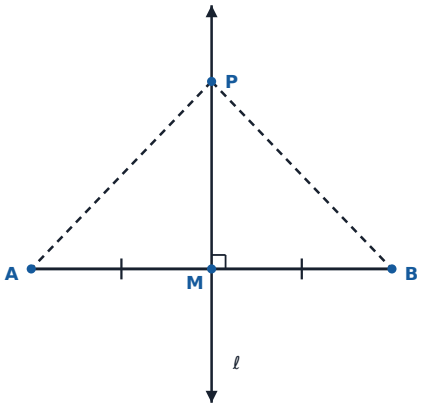
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

652

QUESTION ID

S19-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

PA = PB

GRADING NOTE REQUIRED WORK: Use SAS on triangles PMA and PMB.

Any valid proof (paragraph, two-column, or flow). Essential: AM = BM (M is the midpoint); ∠PMA ≅ ∠PMB (both right angles, $\ell \perp$ AB); PM = PM (reflexive); △PMA ≅ △PMB (SAS); PA = PB (corresponding parts). If P = M, PA = MA = MB = PB directly. Alternative: the reflection across ℓ fixes P and swaps A and B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

653

QUESTION ID

S19-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PM ⊥ AB, so P is on the perpendicular bisector of AB.

GRADING NOTE REQUIRED WORK: Use SSS on triangles PMA and PMB, then the linear pair of congruent angles at M.

Any valid proof (paragraph, two-column, or flow). Essential: PA = PB (given), AM = BM (midpoint), PM = PM; △PMA ≅ △PMB (SSS); ∠PMA ≅ ∠PMB; they form a linear pair, so each is 90°; so PM is perpendicular to AB through its midpoint. Invalid: drawing the perpendicular bisector through P and assuming it passes through P (that is the claim).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

The theorem says the points on the perpendicular bisector of AB are EXACTLY the points equidistant from A and B. Which two statements must be proved?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

654

QUESTION ID
S19-M-Q3

TYPE
Short answer

CORRECT ANSWER
(1) Every point on the perpendicular bisector is equidistant from A and B. (2) Every point equidistant from A and B is on the perpendicular bisector.

GRADING NOTE Any equivalent wording of both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

655

QUESTION ID

S19-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

PA = PB = √(a² + y²) for P(0, y); and PA = PB gives (x + a)² = (x − a)², so x = 0.

GRADING NOTE REQUIRED WORK: Use the distance formula in both directions.

Any valid proof (paragraph, two-column, or flow). Essential: both directions. (1) √(a² + y²) both ways. (2) (x + a)² + y² = (x − a)² + y² simplifies to 4ax = 0, and a ≠ 0, so x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A student proves only that every point on the perpendicular bisector of AB is equidistant from A and B. What is missing?

A. A proof that the bisector is a line

B. Nothing; one direction proves "exactly"

C. The converse: every point equidistant from A and B is on the perpendicular bisector

D. A measurement of PA and PB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

656

QUESTION ID

S19-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The converse: every point equidistant from A and B is on the perpendicular bisector

GRADING NOTE "Exactly those points" needs both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

658

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

659

QUESTION ID

S19-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is its definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. The perpendicular bisector of AB

D. A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

660

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A bisector of AB that is not the perpendicular bisector

GRADING NOTE It bisects the segment but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

661

QUESTION ID

S19-B1-Q4

TYPE

Short answer

CORRECT ANSWER

No. M is not the midpoint (3 ≠ 5).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

662

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Exactly one line is perpendicular to AB at its midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In △PMA and △PMB, PM is shared, AM = BM, and ∠PMA ≅ ∠PMB. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

664

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In △PMA and △PMB, PA = PB, AM = BM, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

665

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. ASA

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

666

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. SSS

GRADING NOTE Three pairs of congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

667

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

668

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two sides and a non-included angle do not always determine a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

670

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠PMB

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

671

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

672

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

△XYZ ≅ △XWZ. Which pair of segments must be congruent?

A. YZ and XW

B. YZ and WZ

C. XY and XZ

D. XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

673

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. YZ and WZ

GRADING NOTE Y ↔ W and Z ↔ Z.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

674

QUESTION ID

S19-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary; they are congruent only when both are 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

676

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = −x + 7

GRADING NOTE Any equivalent equation. Midpoint (4, 3); slope of AB is 1, so the bisector has slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

677

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(7, 0)

GRADING NOTE Form required: an ordered pair. It is where y = −x + 7 meets y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

678

QUESTION ID
S19-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Distances are $\sqrt{32} \approx 5.66$ and $\sqrt{20} \approx 4.47$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

679

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 7)

GRADING NOTE Points on x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

680

QUESTION ID
S19-F1-Q5

TYPE
Short answer

CORRECT ANSWER
The vertical line x = 1.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

682

QUESTION ID

S19-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three perpendicular bisectors meet at O, and OA = OB = OC.

GRADING NOTE REQUIRED WORK: Let the bisectors of AB and BC meet at O; use the theorem and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of AB and BC are not parallel (AB and BC are not parallel), so they meet at O; OA = OB and OB = OC (theorem); so OA = OC; by the converse O is on the bisector of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

683

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 4)

GRADING NOTE Form required: an ordered pair. Bisectors x = 3 and y = 4 meet at (3, 4); it is 5 from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

684

QUESTION ID

S19-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For an obtuse triangle it is outside; for a right triangle it is on the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns are to share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

685

QUESTION ID

S19-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Where the perpendicular bisectors of the triangle's sides meet

GRADING NOTE Points equidistant from two towns are on the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

686

QUESTION ID

S19-F2-Q5

TYPE

Short answer

CORRECT ANSWER

OA = OB and OB = OC give OA = OC, and by the converse O is on the perpendicular bisector of AC.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

SET 20 • HSG-CO.C.10

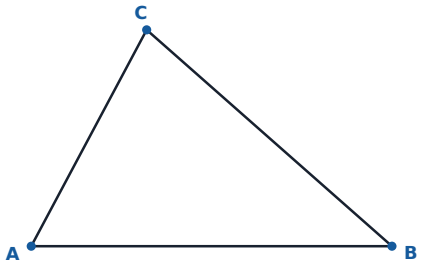
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that
m∠A + m∠B + m∠C = 180° in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

689

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: Name the two new angles at C, use alternate interior angles with AB, and the straight angle at C.

Any valid proof (paragraph, two-column, or flow). Essential: (1) through C there is exactly one line parallel to AB (parallel postulate); (2) the angles it makes with CA and CB are congruent to ∠A and ∠B (alternate interior angles, transversals CA and CB); (3) those two angles and ∠ACB form a straight angle, 180°; (4) substitute. Invalid: measuring angles; citing the exterior angle theorem (it depends on this result).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

690

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠P + m∠Q + m∠R = 180°

GRADING NOTE REQUIRED WORK: Same strategy with vertex P: alternate interior angles and a straight angle at P.

Any valid proof (paragraph, two-column, or flow). Essential: unique parallel through P; the two angles at P next to ∠QPR are congruent to ∠Q and ∠R (alternate interior); the three angles at P make 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

691

QUESTION ID

S20-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠ACB = 180°

GRADING NOTE REQUIRED WORK: Use alternate interior angles (∠A and ∠ACE) and corresponding angles (∠B and ∠ECD), then the straight angle BCD.

Any valid proof (paragraph, two-column, or flow). Essential: ∠ACE ≅ ∠A (alternate interior, transversal AC); ∠ECD ≅ ∠B (corresponding, transversal BD); ∠ACB + ∠ACE + ∠ECD = 180° (straight angle at C).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

A proof of the triangle angle sum (line ℓ through C parallel to AB, making $\angle 1$ next to CA and $\angle 2$ next to CB):

1. $m\angle 1 + m\angle ACB + m\angle 2 = 180°$ (?)
2. $\angle 1 \cong \angle A$ and $\angle 2 \cong \angle B$ (?)
3. $m\angle A + m\angle ACB + m\angle B = 180°$ (?)

Give the three reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

692

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

1: the angles form a straight angle (angle addition); 2: alternate interior angles are congruent ($\ell \parallel$ AB); 3: substitution.

GRADING NOTE Any equivalent names for all three reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

A student proves the angle sum theorem using "an exterior angle equals the sum of the two remote interior angles." What is wrong?

A. Exterior angles do not exist in triangles

B. Nothing is wrong

C. That theorem is usually proved FROM the angle sum, so the argument is circular

D. The exterior angle theorem is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

693

QUESTION ID

S20-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. That theorem is usually proved FROM the angle sum, so the argument is circular

GRADING NOTE Using a result that depends on the theorem being proved is circular reasoning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

695

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

696

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

697

QUESTION ID

S20-B1-Q3

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form. 6x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

698

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Its alternate interior angle

GRADING NOTE Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

699

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two angles over 90° would already add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

701

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

702

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They make a straight angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

∠1, ∠2, and ∠3 form a straight angle, with ∠1 = ∠3 = 50°. Find ∠2.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

703

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 80°

GRADING NOTE 180 − 100 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

704

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form. 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

705

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 180° but can be any pair, such as 30° and 150°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

707

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

58°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

708

QUESTION ID
S20-B3-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the theorem proved in an earlier set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Line ℓ through C is parallel to AB. Which angle at C is congruent to $\angle A$?

A. The angle between CA and CB

B. The angle between ℓ and CB

C. The angle between ℓ and CA

D. $\angle ACB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

709

QUESTION ID

S20-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The angle between ℓ and CA

GRADING NOTE CA is a transversal of ℓ and AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

710

QUESTION ID

S20-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The parallel postulate: through a point not on a line there is exactly one parallel line.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

711

QUESTION ID

S20-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that m∠ACD = m∠A + m∠B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

713

QUESTION ID

S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACD = m∠A + m∠B

GRADING NOTE

REQUIRED WORK: Use the linear pair at C and the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ACD + m∠ACB = 180° (linear pair); m∠A + m∠B + m∠ACB = 180° (angle sum); substitute and subtract m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

714

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Remote interior: 77°; adjacent interior: 55°.

GRADING NOTE Form required: both angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

715

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exterior angle at an obtuse vertex is acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

716

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 25

GRADING NOTE 5x + 10 = 4x + 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

717

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Each exterior angle is 180° minus its interior angle, so the sum is 540° − 180° = 360°.

GRADING NOTE Any valid argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to (n − 2) · 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

719

QUESTION ID

S20-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

(n − 2) · 180°

GRADING NOTE REQUIRED WORK: Draw the diagonals from one vertex to divide the polygon into triangles and use the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: from one vertex there are n − 3 diagonals, making n − 2 triangles; their angles together make up exactly the polygon's angles; each triangle contributes 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

720

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

135°

GRADING NOTE Any equivalent form. $(8 - 2) \cdot 180 \div 8 = 135$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

721

QUESTION ID

S20-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Each exterior angle is 24°; 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

722

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(6 - 2) \cdot 180 = 720°$; 1,080° is the sum for an octagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

723

QUESTION ID

S20-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 115

GRADING NOTE 540 − 425 = 115.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

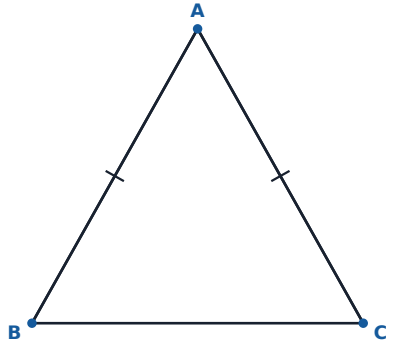
SET 21 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

726

QUESTION ID

S21-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △ACD by SAS, then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC (given); ∠BAD ≅ ∠CAD (definition of angle bisector); AD = AD; △ABD ≅ △ACD (SAS); ∠B ≅ ∠C (corresponding angles). Invalid: "∠B and ∠C look equal"; using the fact that AD ⊥ BC (not yet known).

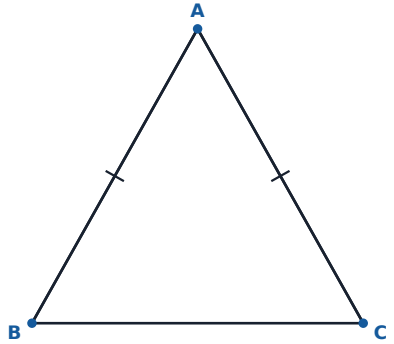
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

727

QUESTION ID

S21-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE

REQUIRED WORK: Prove △ABM ≅ △ACM by SSS.

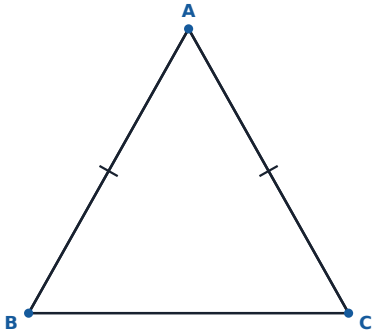
Any valid proof (paragraph, two-column, or flow). Essential: AB = AC; BM = CM (midpoint); AM = AM; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

728

QUESTION ID

S21-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The reflection maps ∠ABC onto ∠ACB, so ∠B ≅ ∠C.

GRADING NOTE REQUIRED WORK: Explain where the reflection sends ray AB, point B, and point C.

Any valid explanation. Essential: the reflection across the bisector of ∠A swaps rays AB and AC (the bisector makes equal angles with them); B goes to the point of ray AC at distance AB = AC, which is C; so B ↔ C and the triangle maps onto itself with ∠B ↔ ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

729

QUESTION ID

S21-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

All three angles are congruent (each is 60°).

GRADING NOTE REQUIRED WORK: Apply the isosceles base angle theorem twice, with two different pairs of equal sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC gives ∠B ≅ ∠C; AB = BC gives ∠A ≅ ∠C; so all three are congruent (and each is 60° by the angle sum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

In the proof that base angles of isosceles triangle ABC (AB = AC) are congruent, the bisector of ∠A meets BC at D. Which criterion proves △ABD ≅ △ACD?

A. SAS

B. SSS

C. AAS

D. ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

730

QUESTION ID

S21-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. SAS

GRADING NOTE AB = AC, ∠BAD ≅ ∠CAD, and AD is shared: two sides and the included angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

732

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM.
Which criterion shows △ABM ≅ △ACM?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

733

QUESTION ID
S21-B1-Q2

TYPE
Short answer

CORRECT ANSWER
SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

734

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

735

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two sides and a non-included angle

GRADING NOTE SSA is not a criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If △ABD ≅ △ACD, then ∠ABD ≅ ∠ACD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

736

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

738

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

739

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

740

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

741

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. m∠BAD = m∠DAC

GRADING NOTE Equal angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of ∠A of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

742

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when AB = AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

744

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Base QR; base angles ∠Q and ∠R

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

745

QUESTION ID

S21-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition used in this course.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

746

QUESTION ID

S21-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Sides 4, 5, 6

GRADING NOTE No two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

747

QUESTION ID

S21-B3-Q4

TYPE

Short answer

CORRECT ANSWER

∠X

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

748

QUESTION ID

S21-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It has at least two congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, ∠B ≅ ∠C. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

750

QUESTION ID

S21-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = AC

GRADING NOTE REQUIRED WORK: Draw the bisector of ∠A meeting BC at D and prove △ABD ≅ △ACD by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (given); ∠BAD ≅ ∠CAD (bisector); AD = AD; AAS; AB = AC. Invalid: using AB = AC to choose the auxiliary line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

751

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4; PQ = 14

GRADING NOTE Form required: x and PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

752

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the converse of the isosceles triangle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has m∠A = 50° and m∠B = 65°. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

753

QUESTION ID
S21-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. AB and AC

GRADING NOTE m∠C = 65° = m∠B, so the sides opposite them, AC and AB, are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

754

QUESTION ID

S21-F1-Q5

TYPE

Answer with required work

CORRECT ANSWER

All three sides are congruent.

GRADING NOTE

REQUIRED WORK: Apply the converse twice.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠B gives BC = AC; ∠B ≅ ∠C gives AC = AB; so all three sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

756

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

71°

GRADING NOTE Any equivalent form. (180 − 38) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

757

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. Triangles ABD and ACD are isosceles, so ∠BAD = ∠B and ∠CAD = ∠C; then 2(∠B + ∠C) = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

758

QUESTION ID

S21-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

AD = AE

GRADING NOTE

REQUIRED WORK: Use the base angles and SAS on triangles ABD and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (base angles); AB = AC; BD = CE; △ABD ≅ △ACE (SAS); AD = AE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

759

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (180 − 100) ÷ 2 = 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

760

QUESTION ID

S21-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 44°

GRADING NOTE ∠PQR = 68° = ∠R, so ∠P = 180 − 136 = 44.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

763

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

D(b, c), E(a + b, c); DE is horizontal like AB, and DE = a = ½(2a) = ½AB.

GRADING NOTE REQUIRED WORK: Find D and E with the midpoint formula, then compare slopes and lengths.

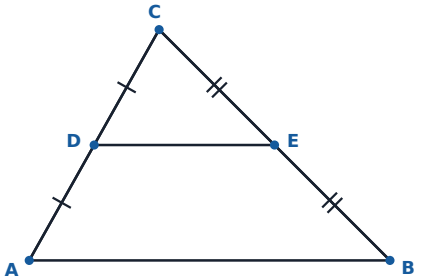
Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; DE has slope 0 (same y) and AB has slope 0, so DE ∥ AB (and DE ≠ AB since c ≠ 0); DE = |a + b − b| = a and AB = 2a. Invalid: checking one numerical triangle only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

764

QUESTION ID

S22-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ AB and DE = ½AB

GRADING NOTE REQUIRED WORK: Prove △DEC ≅ △FEB (SAS), deduce BF = DC = AD and BF ∥ AD, so ABFD is a parallelogram; then use its opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: CE = BE, DE = FE, vertical angles at E, so △DEC ≅ △FEB (SAS); BF = DC = AD; ∠DCE ≅ ∠FBE gives BF ∥ AC (alternate interior converse); one pair of opposite sides of ABFD is parallel and congruent, so ABFD is a parallelogram; so DF ∥ AB and DF = AB; DE = ½DF = ½AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

In the coordinate proof of the midsegment theorem, the vertices are A(0, 0), B(2a, 0), C(2b, 2c). Why are the coordinates written with 2a, 2b, and 2c?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

765

QUESTION ID

S22-M-Q3

TYPE

Short answer

CORRECT ANSWER

So the midpoints have whole-letter coordinates (a, b, c) without fractions, which keeps the algebra simple; any triangle can still be placed this way.

GRADING NOTE Any equivalent explanation that mentions avoiding fractions in the midpoint formula and that the placement is general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

A proof of the midsegment theorem with A(0, 0), B(2a, 0), C(2b, 2c):

1. D = (b, c), E = (a + b, c) (?)
2. slope DE = 0 = slope AB (?)
3. DE ∥ AB (?)
4. DE = a and AB = 2a, so DE = ½AB (?)

Give the four reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

766

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

1: midpoint formula; 2: slope formula (equal y-coordinates); 3: lines with equal slopes are parallel; 4: distance formula (horizontal distances).

GRADING NOTE Any equivalent names for all four reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

A student checks the triangle A(0, 0), B(6, 0), C(2, 4), finds the midsegment is parallel to AB and 3 units long, and concludes the theorem is proved. What is the flaw?

A. One example cannot prove a statement about every triangle

B. The midsegment should be 6 units long

C. Midpoints cannot be found with coordinates

D. The midsegment is not parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

767

QUESTION ID

S22-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. One example cannot prove a statement about every triangle

GRADING NOTE A general proof must work for every triangle, for example with variable coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

769

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (−3, 1) to (5, −7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

770

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

771

QUESTION ID

S22-B1-Q3

TYPE

Short answer

CORRECT ANSWER

(a, c)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

772

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ((1 + 7) ÷ 2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

773

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

775

GRADING NOTE Any equivalent form. (8 − 2) ÷ (4 − 1).

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (b, c) and (a + b, c).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

776

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. The y-coordinates are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

777

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same steepness, different positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

778

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. From (1, 3) to (5, 5)

GRADING NOTE Both have slope 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

779

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Horizontal has slope 0; vertical is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

781

QUESTION ID

S22-B3-Q1

TYPE

Short answer

CORRECT ANSWER

a (that is, |a|, with a > 0)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

782

QUESTION ID

S22-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from (−2, 5) to (4, 5).

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

783

QUESTION ID

S22-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Horizontal: |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

784

QUESTION ID

S22-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(9 + 16)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

785

QUESTION ID

S22-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

787

QUESTION ID

S22-B4-Q1

TYPE

Short answer

CORRECT ANSWER

$\ell \parallel m$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

788

QUESTION ID
S22-B4-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Converse of the alternate interior angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In △ABC, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that EF = DE, and △DEC ≅ △FEB. Which fact shows BF ∥ AC?

A. DE = EF

B. ∠DCE ≅ ∠FBE are alternate interior angles

C. BF = DC

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

789

QUESTION ID
S22-B4-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. ∠DCE ≅ ∠FBE are alternate interior angles

GRADING NOTE Congruent alternate interior angles (transversal BC) give parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

790

QUESTION ID

S22-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

791

QUESTION ID

S22-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Vertical angles are congruent for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

793

QUESTION ID

S22-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ ∥ AC ∥ SR and PQ = ½AC = SR, so PQRS is a parallelogram.

GRADING NOTE REQUIRED WORK: Draw diagonal AC and apply the midsegment theorem in triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ABC, PQ ∥ AC and PQ = ½AC; in △ADC, SR ∥ AC and SR = ½AC; so PQ ∥ SR and PQ = SR; a quadrilateral with one pair of opposite sides parallel and congruent is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

794

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. Its sides are ½AC, ½BD, ½AC, ½BD: 5 + 7 + 5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

795

QUESTION ID

S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is Varignon's theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

796

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. rectangle

GRADING NOTE Its sides are parallel to the diagonals, so adjacent sides are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

797

QUESTION ID

S22-F1-Q5

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. Each midsegment is half the parallel side: 6 + 8 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

799

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each side of DEF is half of a side of ABC (midsegment theorem), so the triangles are similar by SSS similarity with scale factor ½.

GRADING NOTE Any equivalent explanation with the factor ½ (or 2 from DEF to ABC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

800

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form. Scale factor ½ gives area factor ¼; also the 4 small triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5